

SUSPENSION

02
SECTION

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02-10 GENERAL PROCEDURES

GENERAL PROCEDURES (SUSPENSION)	02-10-1
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GENERAL PROCEDURES (SUSPENSION)

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Wheel And Tire Installation

1. When installing the wheels and tires, tighten the wheel nuts in a criss-cross pattern to the following tightening torque.

Tightening torque

88.2—117.6 N·m {9.00—11.99 kgf·m, 65.06—86.73 ft·lbf}

Suspension Links Removal/Installation

1. For the joint sections with rubber bushings, raise the vehicle using a lift, and then temporarily tighten the installation bolts and nuts. Lower the vehicle to the ground and tighten them completely with the specified torque.

Connector Disconnection

1. Disconnect the negative battery cable before performing any work that requires handling of connectors.

02

(1)

(2)

(3)

02-11 WHEEL ALIGNMENT

VEHICLE HEIGHT

ADJUSTMENT [4x2] 02-11-1

VEHICLE HEIGHT

ADJUSTMENT [4x4] 02-11-2

WHEEL ALIGNMENT

PRE-INSPECTION 02-11-2

FRONT WHEEL ALIGNMENT [4x2] 02-11-3

FRONT WHEEL ALIGNMENT [4x4] 02-11-5

VEHICLE HEIGHT ADJUSTMENT [4x2]

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Caution

- If the vehicle height on a winch or towbar (bullbar or similar accessory) equipped vehicle is adjusted, it is possible that the vehicle height will drop below specification again or the tires will be subject to uneven wear. Without adjusting the vehicle height, adjust only the difference between the left and right sides and perform a wheel alignment to specification.

1. Place the vehicle on level ground.
2. Inspect the front and rear tire pressure and adjust it as necessary.
3. Measure the distance from the center of each front wheel to the fender brim.

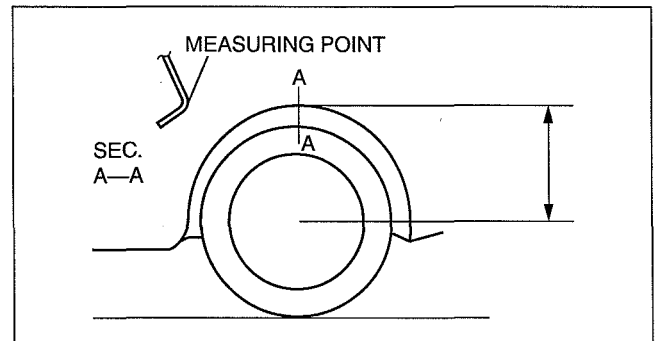
Standard vehicle height

Except Hi-Rider: 416—456 mm {16.4—17.9 in}

Hi-Rider: 512—552 mm {20.2—21.7 in}

Left/right difference

10 mm {0.39 in} max.

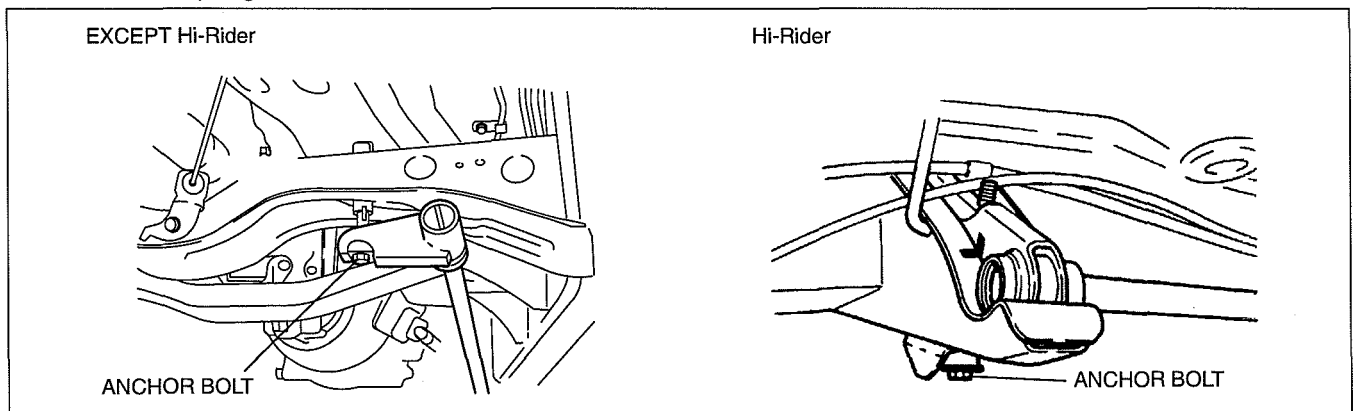


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Caution

- Adjust the vehicle height at the left and right wheels at the same time. Otherwise, the vehicle height may not be adjusted correctly.

4. If the difference between left and right is not within the specification, adjust the vehicle height by turning the torsion bar spring anchor bolt.



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5. Move the vehicle forward and back to seat each part.
6. Verify that the vehicle height is within the standard.
 - If not, readjust the vehicle height.

WHEEL ALIGNMENT

VEHICLE HEIGHT ADJUSTMENT [4x4]

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Caution

- If the vehicle height on a winch or towbar (bullbar or similar accessory) equipped vehicle is adjusted, it is possible that the vehicle height will drop below specification again or the tires will be subject to uneven wear. Without adjusting the vehicle height, adjust only the difference between the left and right sides and perform a wheel alignment to specification.

1. Place the vehicle on level ground.
2. Inspect the front and rear tire pressure and adjust it as necessary.
3. Measure the distance from the center of each front wheel to the fender brim.

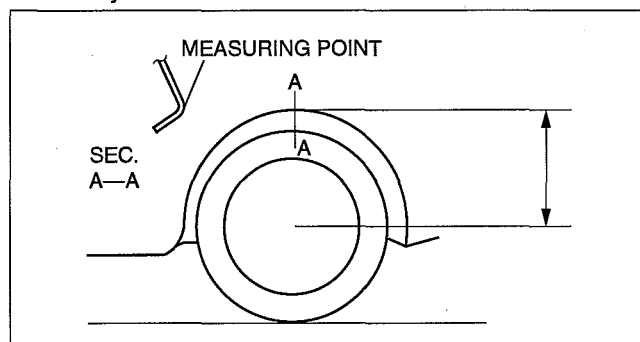
Standard vehicle height

Regular cab and Stretch cab (with Rear Access System): 512—552 mm {20.2—21.7 in}

Double cab: 502—542 mm {19.8—21.3 in}

Left/right difference

10 mm {0.39 in} max.



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Caution

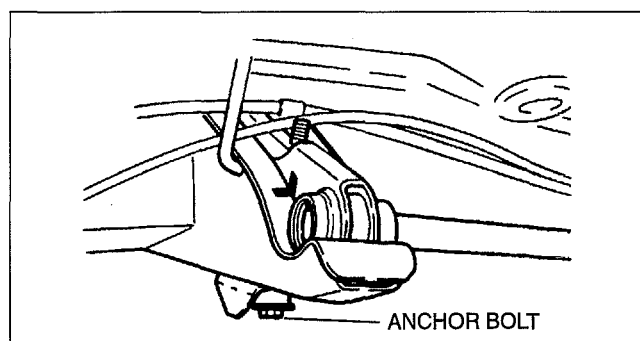
- Adjust the vehicle height at the left and right wheels at the same time. Otherwise, the vehicle height may not be adjusted correctly.

4. If the difference between left and right is not within the specification, adjust the vehicle height by turning the torsion bar spring anchor bolt.

Note

- The vehicle height increases/decreases 3.3 mm {0.13 in} per anchor bolt rotation.

5. Move the vehicle forward and back to seat each part.
6. Verify that the vehicle height is within the standard.
 - If not, readjust the vehicle height.



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WHEEL ALIGNMENT PRE-INSPECTION

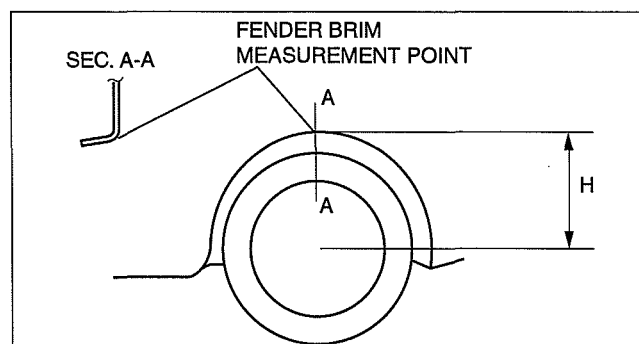
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1. Inspect the tire pressure.
 - Adjust to the recommended pressure if necessary. (See 02-50-1 SUSPENSION TECHNICAL DATA.)
2. Inspect the wheel bearing play.
 - Correct if necessary. (See 03-11-3 WHEEL HUB, STEERING KNUCKLE INSPECTION [4x2 (EXCEPT Hi-Rider)].) (See 03-11-4 WHEEL HUB, STEERING KNUCKLE INSPECTION [Hi-Rider, 4x4].) (See 03-12-1 REAR AXLE INSPECTION.)
3. Inspect the wheel runout.
 - Correct if necessary. (See 02-50-1 SUSPENSION TECHNICAL DATA.)
4. Inspect the ball joints and steering linkage for excessive looseness.
5. Park the vehicle on a level ground, in an unloaded condition*, and with the wheels straight forward.
Unloaded vehicle.....Fuel tank is full. Engine coolant and engine oil are at specified level. Jack and tools are in designated position.

WHEEL ALIGNMENT

6. Measure height H from the center of the wheel to the fender brim.
7. Verify that the difference between the left and right dimension H is within the specification.
 - If it exceeds the specification, repeat the Step 2—6.

Standard
10 mm {0.39 in} or less



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FRONT WHEEL ALIGNMENT [4x2]

Front wheel alignment [4x2] (Unloaded)

Item			Specifications			
			Regular cab	Double cab	Hi-Rider (Stretch cab (with Rear Access System))	Hi-Rider (Double cab)
Total toe-in	Tire [Tolerance ± 3 {0.12}]	(mm {in})	5 {0.19}			
Maximum steering angle	Inner		33°00'—37°00'		31°30'—35°30'	
	Outer		30°00'—35°00'		27°00'—32°00'	
Camber angle [Tolerance $\pm 1^\circ$]			0°35'		0°44'	0°45'
Caster angle [Tolerance $\pm 1^\circ$]			1°56'	1°54'	2°07'	2°06'
Steering axis inclination (reference value)			8°25'		10°37'	10°35'

Note

- Unloaded vehicle: Fuel tank is full. Engine coolant and engine oil are at specified level. Jack and tools are in designated position.
- Difference between the left and right dimension for camber is within 30'.
- Difference between the left and right dimension for caster is within 45'.

Maximum Steering Angle Adjustment

1. Loosen the adjusting bolt locknut.
2. Turn the adjusting bolt to provide the correct turning angle.

Standard steering angle

Except Hi-Rider

Inner: 33°00'—37°00'

Outer: 30°00'—35°00'

Hi-Rider

Inner: 31°30'—35°30'

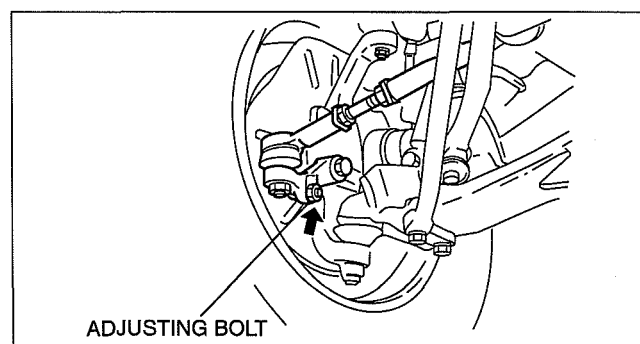
Outer: 27°00'—32°00'

3. After adjustment, tighten the locknut to the specified torque.

Tightening torque

39—59 N·m {4.0—6.0 kgf·m, 29—43 ft·lbf}

4. After adjusting the steering angle, always inspect and adjust the toe angle. (See 02-11-4 Total Toe-in Adjustment.)



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WHEEL ALIGNMENT

Caster And Camber Adjustment

1. Loosen the upper arm shaft bolt and insert or remove adjustment shims to provide the correct angle.

No.	Thickness (mm {in})
1	1.0 {0.039}
2	1.6 {0.063}
3	2.0 {0.079}
4	3.2 {0.126}
5	4.0 {0.157}
6	0.6 {0.024}
7	1.0 {0.039}
8	1.6 {0.063}
9	2.0 {0.079}
10	3.2 {0.126}
11	4.0 {0.157}

Note

- Shims No.1—5 are used on either side. Do not use more than two.
- Shims No.6—11 are used at the front and rear of either sides. Do not use more than one.
- Camber (Shims No.1—5): A change of shim thickness of 1 mm {0.039 in} results as follows:

Shim thickness	Variation/1 mm {0.039 in}
Add	Negative 15'
Reduce	Positive 15'

- Caster (Shims No.6—11): A change of shim thickness (at front or rear only) of 1 mm {0.039 in} results as follows:

	Shim thickness	Variation/1 mm {0.039 in}
Front side	Add	Increase 30'
	Reduce	Decrease 30'
Rear side	Add	Decrease 30'
	Reduce	Increase 30'

2. After adjustment, tighten the upper arm shaft bolt to the specified torque.

Tightening torque

150—172 N·m {15.3—17.5 kgf·m, 111—126 ft·lbf}

Total Toe-in Adjustment

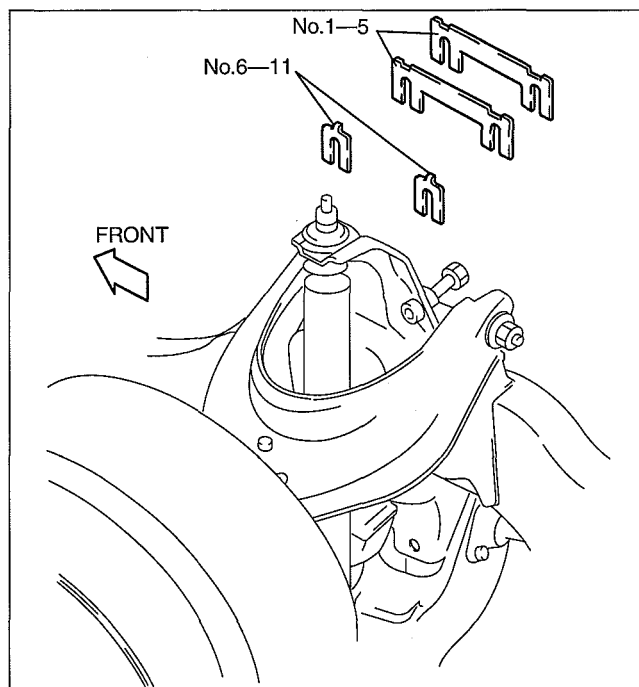
1. Loosen the locknut of the tie-rod end.
2. Turn the tie rods by the same amount in the opposite direction.

Total Toe-in Standard

2—8 mm {0.08—0.31 in}

Note

- One turn of the tie rod (both sides) changes the toe-in by about **approx. 30 mm {1.18 in}**.
- The left and right tie rods are both right threaded. To increase the toe-in, turn the right tie rod toward the front of the vehicle and the left tie rod equally toward the rear.



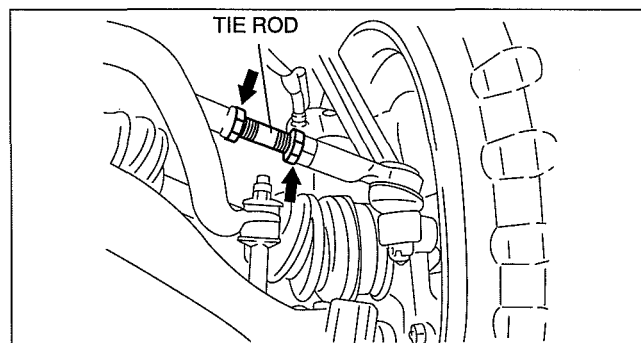
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WHEEL ALIGNMENT

3. Tighten the tie-rod end locknuts.

Tightening torque

68.6—78.5 N·m {7.00—8.00 kgf·m, 50.6—57.8 ft·lbf}



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FRONT WHEEL ALIGNMENT [4x4]

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Front wheel alignment [4x4] (Unloaded)

Item			Specifications	
			Regular cab and Stretch cab (with Rear Access System)	Double cab
Total toe-in	Tire [Tolerance ± 3 {0.12}]	(mm {in})	6 {0.23}	
Maximum steering angle		Inner	31°30'—35°30'	
		Outer	27°00'—32°00'	
Camber angle (reference value) [Tolerance $\pm 1^\circ$]			0°44'	0°45'
Caster angle (reference value) [Tolerance $\pm 1^\circ$]			2°07'	2°06'
Steering axis inclination (reference value)			10°37'	10°35'

Note

- Unloaded vehicle: Fuel tank is full. Engine coolant and engine oil are at specified level. Jack and tools are in designated position.
- Difference between the left and right dimension for camber is within 30'.
- Difference between the left and right dimension for caster is within 45'.

Maximum Steering Angle Adjustment

1. Remove the cap.
2. Loosen the adjusting bolt locknut.
3. Turn the adjusting bolt to provide the correct turning angle.

Standard steering angle

Inner: 31°30'—35°30'

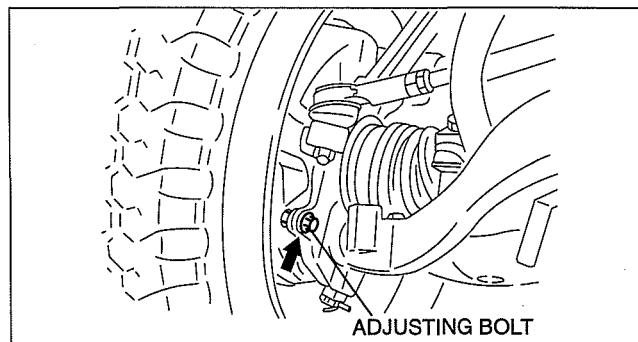
Outer: 27°00'—32°00'

4. After adjustment, tighten the locknut to the specified torque.

Tightening torque

39—59 N·m {4.0—6.0 kgf·m, 29—43 ft·lbf}

5. Install the cap.



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WHEEL ALIGNMENT

Caster and Camber Adjustment

1. Loosen the upper arm shaft bolt and insert or remove adjustment shims to provide the correct angle.

Adjusting shim

No.	Thickness (mm {in})
1	1.0 {0.039}
2	1.6 {0.063}
3	2.0 {0.079}
4	3.2 {0.126}
5	4.0 {0.157}
6	0.6 {0.024}
7	1.0 {0.039}
8	1.6 {0.063}
9	2.0 {0.079}
10	3.2 {0.126}
11	4.0 {0.157}

Note

- Shims No.1—5 are used on either side. Do not use more than two.
- Shims No.6—11 are used at the front and rear of either sides. Do not use more than one.
- Camber (Shims No.1—5): A change of shim thickness of **1 mm {0.039 in}** results as follows:

Shim thickness	Variation/1 mm {0.039 in}
Add	Negative 15'
Reduce	Positive 15'

- Caster (Shims No.6—11): A change of shim thickness (at front or rear only) of **1 mm {0.039 in}** results as follows:

	Shim thickness	Variation/1 mm {0.039 in}
Front side	Add	Increase 30'
	Reduce	Decrease 30'
Rear side	Add	Decrease 30'
	Reduce	Increase 30'

2. After adjustment, tighten the upper arm shaft bolt to the specified torque.

Tightening torque

150—172 N·m {15.3—17.5 kgf·m, 111—126 ft·lbf}

3. Confirm the front wheel alignment as in the specification.
 - If not within the specification, adjust the front wheel alignment again.

Total Toe-in Adjustment

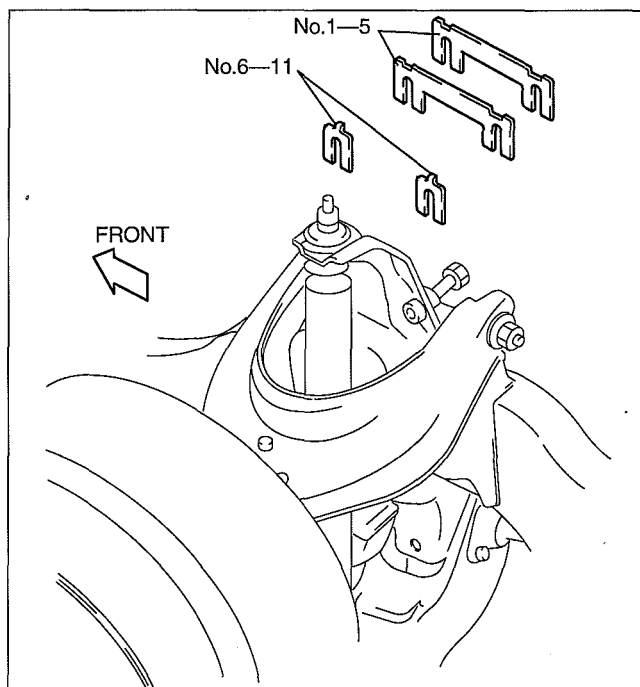
1. Loosen the tie-rod end locknuts.
2. Turn the tie rods by the same amount in the opposite direction.

Total Toe-in Standard

3—9 mm {0.12—0.35 in}

Note

- One turn of the tie rod (both sides) changes the toe-in by about **30 mm {1.18 in}**.
- The left and right tie rods are both right threaded, to increase the toe-in, turn the right tie rod toward the front of the vehicle and the left tie rod equally toward the rear.



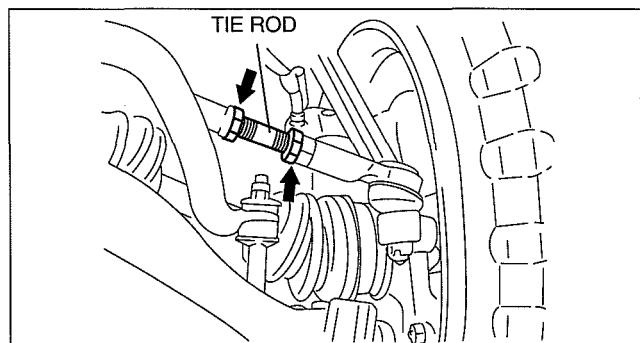
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WHEEL ALIGNMENT

3. Tighten the tie-rod end locknuts.

Tightening torque

**68.6—78.5 N·m {7.00—8.00 kgf·m, 50.6—57.8
ft·lbf}**



AVF7412W002

WHEEL AND TIRES

02-12 WHEEL AND TIRES

WHEEL AND TIRE
SPECIFICATION 02-12-1

WHEEL AND TIRE
SPECIFICATION 02-12-1

WHEEL AND TIRE SPECIFICATION

id0212008001b1

Wheel and tires

Item				Specification			
Wheel	Size			15 × 6 1/2J		15 × 6 1/2J	16 × 7J
	Offset (mm {in})			20 {0.79}		25 {0.98}	10 {0.39}
	Pitch circle diameter (mm {in})			139.7 {5.500}			
	Material			Steel	Aluminum alloy	Steel	Aluminum alloy
Tire	Size			215/70R15C 106/104S		235/75R15 109S	245/70R16 111S
	Air pressure (kPa {(kgf/cm ²), <psi>})	Front	Up to full persons	220 {(2.2), <32>}		210 {(2.1), <30>}	210 {(2.1), <30>}
			Full load				
		Rear	Up to full persons	210 {(2.1), <30>}		210 {(2.1), <30>}	210 {(2.1), <30>}
			Full load	375 {(3.8), <54>}		290 {(2.9), <42>}	270 {(2.7), <39>}
	Remaining tread (mm {in})			1.6 {0.06}			
Wheel and tire	Lug nut tightening torque (N·m {kgf·m, ft·lbf})			88.2—117.6 {9.00—11.99, 65.06—86.73}			
	Wheel and tire runout (mm {in})		Radial direction	1.5 {0.06} max.			
			Lateral direction	2.0 {0.08} max.			
	Wheel imbalance*2 (g {oz})			10 {0.35} max.			

*1 : Total weight exceeds **120 g {4.24 oz}**.

*2 : One balance weight: **60 g {2.12 oz}** max. If the total weight exceeds **120 g {4.24 oz}** on one side, rebalance after moving the tire around on the rim. Do not use three or more balance weights.

WHEEL AND TIRE SPECIFICATION

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Wheel and tires

Item				Specification			
Wheel	Size			15 × 6 1/2J		15 × 6 1/2J	16 × 7J
	Offset (mm {in})			20 {0.79}		25 {0.98}	10 {0.39}
	Pitch circle diameter (mm {in})			139.7 {5.500}			
	Material			Steel	Aluminum alloy	Steel	Aluminum alloy
Tire	Size			215/70R15C 106/104S		235/75R15 109S	245/70R16 111S
	Air pressure (kPa {(kgf/cm ²), <psi>})	Front	No loaded	220 {(2.2), <32>}		210 {(2.1), <30>}	210 {(2.1), <30>}
			Loaded				
		Rear	No loaded	210 {(2.1), <30>}		210 {(2.1), <30>}	210 {(2.1), <30>}
			Loaded	375 {(3.8), <54>}		290 {(2.9), <42>}	270 {(2.7), <39>}
	Remaining tread (mm {in})			1.6 {0.06}			
Wheel and tire	Lug nut tightening torque (N·m {kgf·m, ft·lbf})			88.2—117.6 {9.00—11.99, 65.06—86.73}			
	Wheel and tire runout (mm {in})		Radial direction	1.5 {0.06} max.			
			Lateral direction	2.0 {0.08} max.			
	Wheel imbalance*2 (g {oz})			10 {0.35} max.			

*1 : Total weight exceeds **120 g {4.24 oz}**.

*2 : One balance weight: **60 g {2.12 oz}** max. If the total weight exceeds **120 g {4.24 oz}** on one side, rebalance after moving the tire around on the rim. Do not use three or more balance weights.

02-13 FRONT SUSPENSION

FRONT SUSPENSION LOCATION

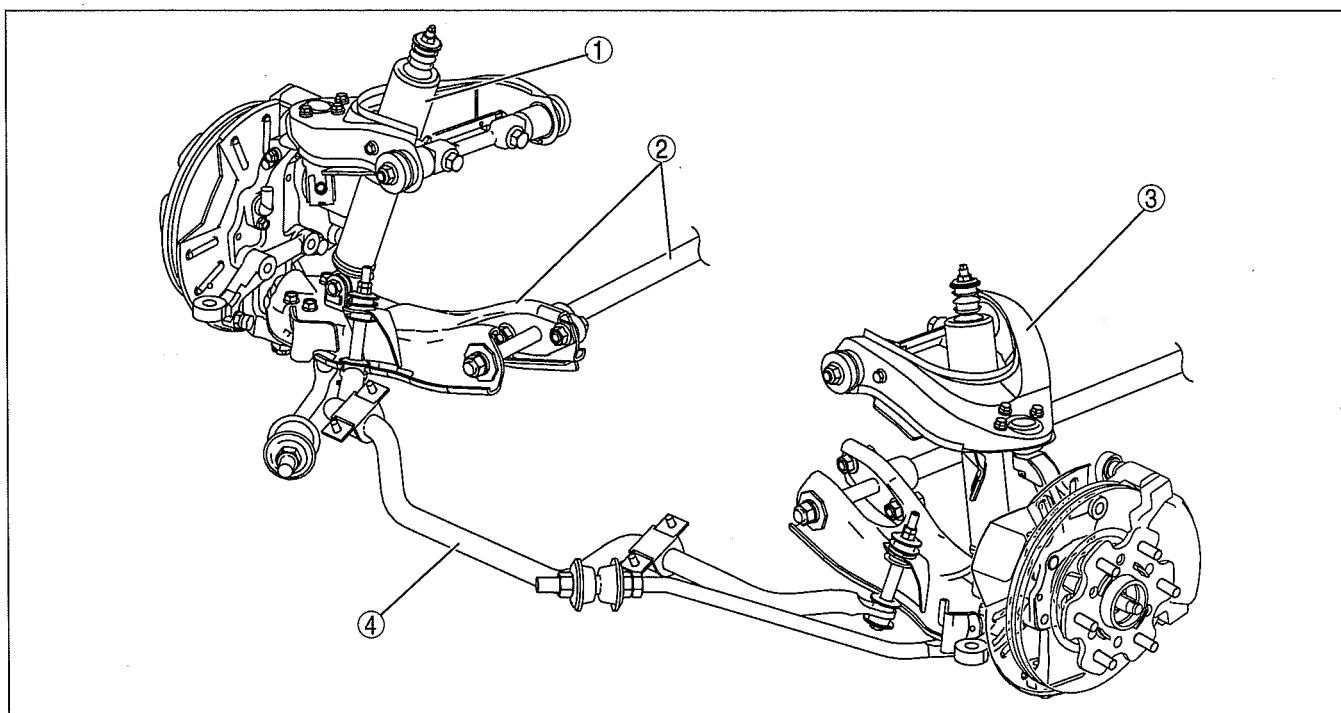
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TORSION BAR SPRING AND LOWER

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FRONT SUSPENSION LOCATION INDEX [4x2 (EXCEPT Hi-Rider)]

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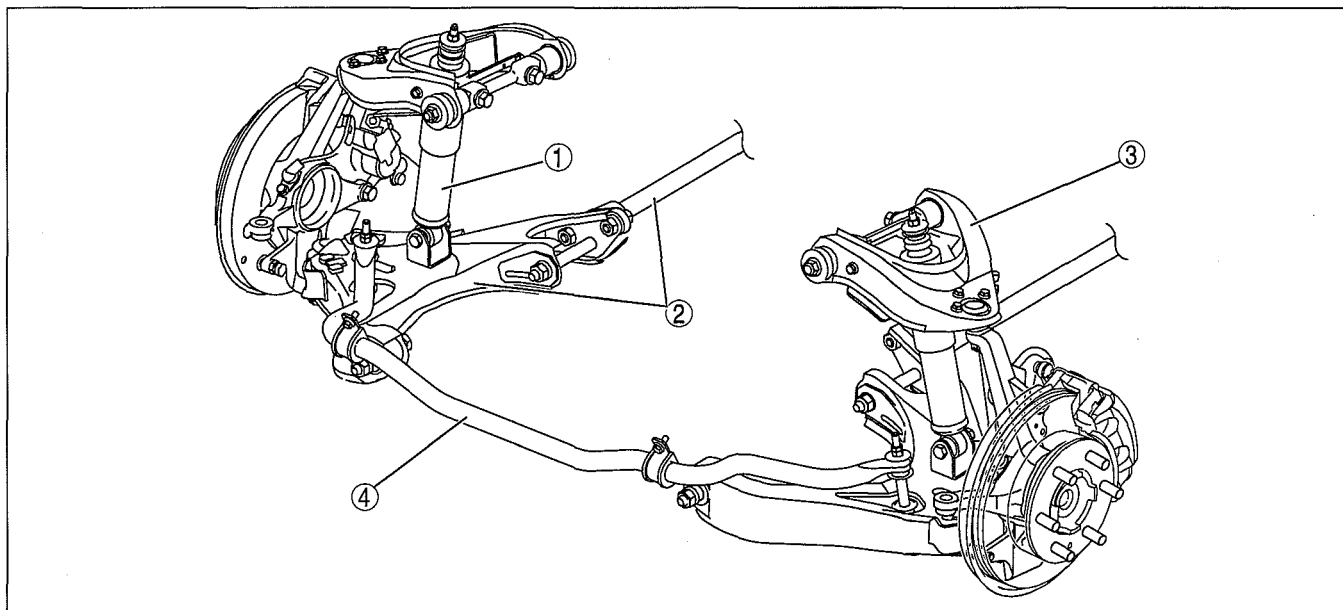
1	Front shock absorber (See 02-13-4 FRONT SHOCK ABSORBER REMOVAL/INSTALLATION.) (See 02-13-4 FRONT SHOCK ABSORBER INSPECTION.) (See 02-13-5 FRONT SHOCK ABSORBER DISPOSAL.)
2	Torsion bar spring, lower arm (See 02-13-5 TORSION BAR SPRING AND LOWER ARM REMOVAL/INSTALLATION [4x2 (EXCEPT Hi-Rider)].) (See 02-13-13 FRONT LOWER ARM INSPECTION.)

3	Front upper arm (See 02-13-14 FRONT UPPER ARM REMOVAL/ INSTALLATION.) (See 02-13-16 FRONT UPPER ARM INSPECTION.)
4	Front stabilizer (See 02-13-16 FRONT STABILIZER AND TENSION ROD REMOVAL/INSTALLATION [4x2 (EXCEPT Hi- Rider)].)

FRONT SUSPENSION

FRONT SUSPENSION LOCATION INDEX [Hi-Rider]

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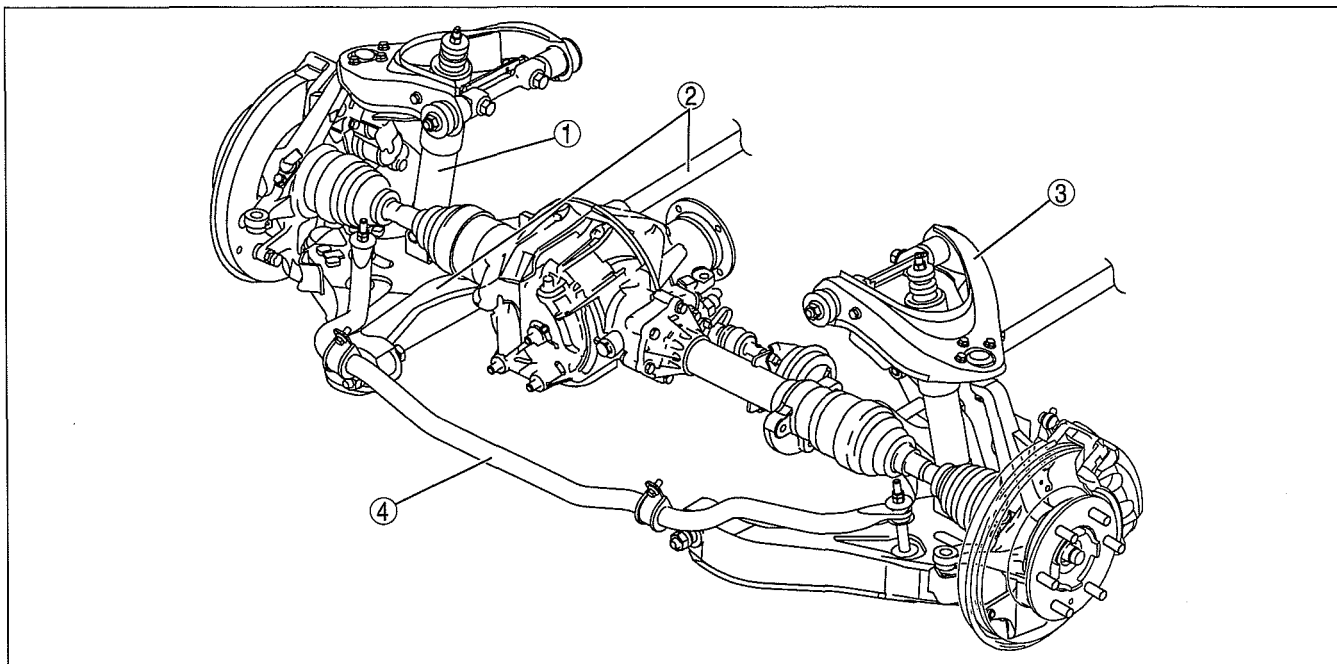
1	Front shock absorber (See 02-13-4 FRONT SHOCK ABSORBER REMOVAL/INSTALLATION.) (See 02-13-4 FRONT SHOCK ABSORBER INSPECTION.) (See 02-13-5 FRONT SHOCK ABSORBER DISPOSAL.)
2	Torsion bar spring, lower arm (See 02-13-9 TORSION BAR SPRING AND LOWER ARM REMOVAL/INSTALLATION [Hi-Rider, 4x4].) (See 02-13-13 FRONT LOWER ARM INSPECTION.)

3	Front upper arm (See 02-13-14 FRONT UPPER ARM REMOVAL/INSTALLATION.) (See 02-13-16 FRONT UPPER ARM INSPECTION.)
4	Front stabilizer (See 02-13-17 FRONT STABILIZER REMOVAL/INSTALLATION [Hi-Rider, 4x4].)

FRONT SUSPENSION

FRONT SUSPENSION LOCATION INDEX [4x4]

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02

DBG213ZWB030

1	Front shock absorber (See 02-13-4 FRONT SHOCK ABSORBER REMOVAL/INSTALLATION.) (See 02-13-4 FRONT SHOCK ABSORBER INSPECTION.) (See 02-13-5 FRONT SHOCK ABSORBER DISPOSAL.)
2	Torsion bar spring, lower arm (See 02-13-9 TORSION BAR SPRING AND LOWER ARM REMOVAL/INSTALLATION [Hi-Rider, 4x4].) (See 02-13-13 FRONT LOWER ARM INSPECTION.)

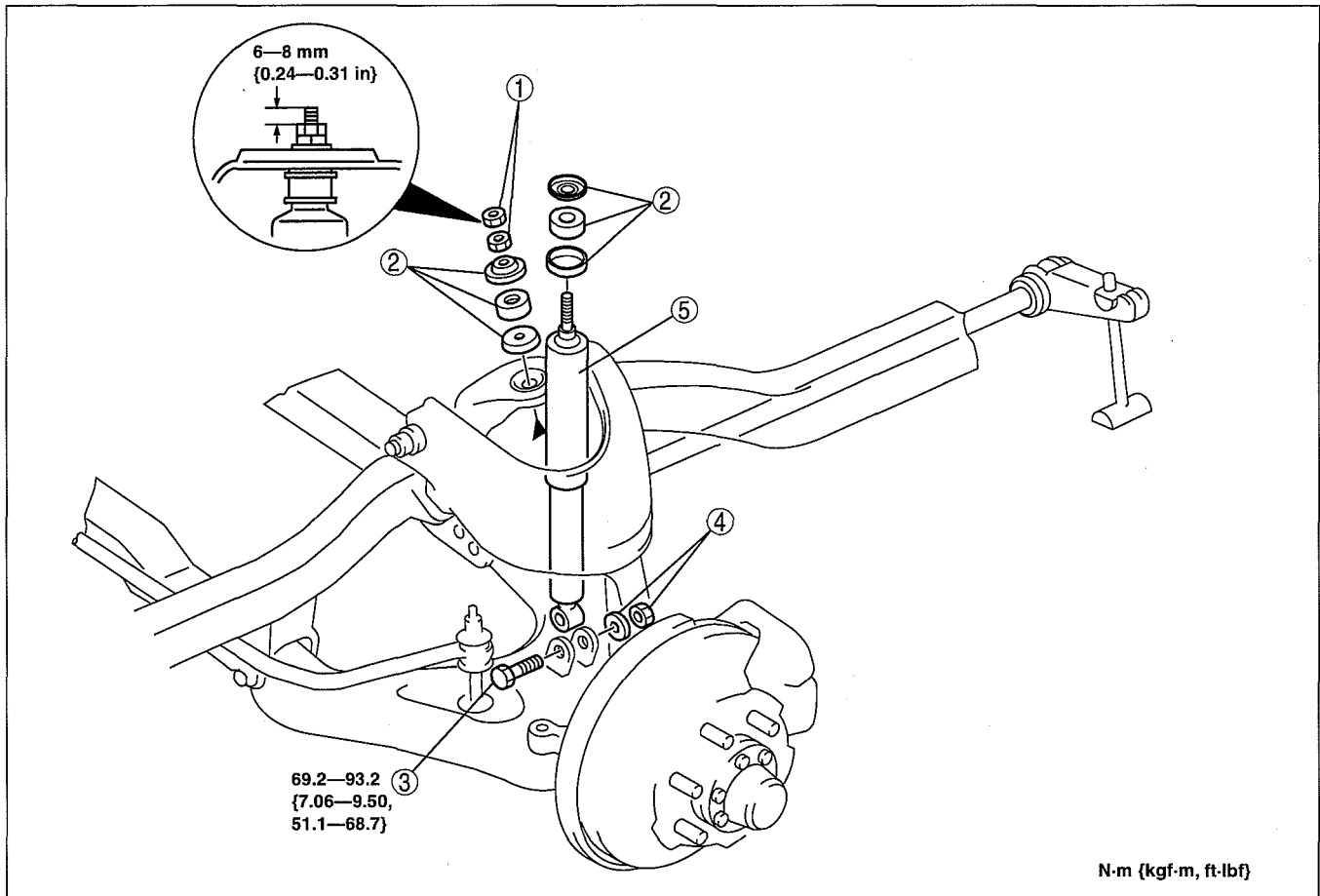
3	Front upper arm (See 02-13-14 FRONT UPPER ARM REMOVAL/ INSTALLATION.) (See 02-13-16 FRONT UPPER ARM INSPECTION.)
4	Front stabilizer (See 02-13-17 FRONT STABILIZER REMOVAL/ INSTALLATION [Hi-Rider, 4x4].)

FRONT SUSPENSION

FRONT SHOCK ABSORBER REMOVAL/INSTALLATION

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1. Remove in the order indicated in the table.
2. Install in the reverse order of removal.



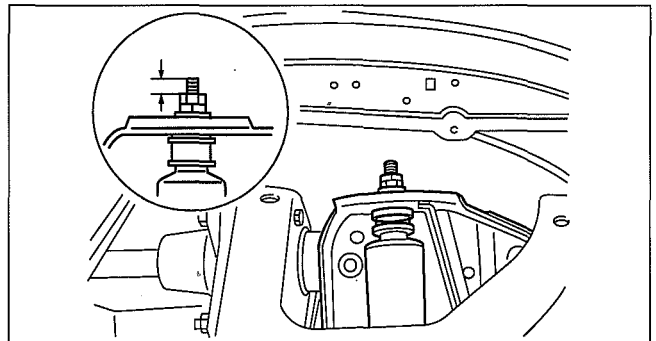
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1	Nut (See 02-13-4 Nut Installation Note.)
2	Retainer, bushing

3	Bolt
4	Washer and nut
5	Front shock absorber

Nut Installation Note

1. Tighten the nut so that the exposed thread of the shock absorber piston is 6-8 mm {0.24-0.31 in}.



DBG0213ZWB00

FRONT SHOCK ABSORBER INSPECTION

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1. Remove the front shock absorber.
2. Inspect for damage and oil leakage.
3. Compress and extend the shock absorber piston rod at least three times at a steady speed. From the fourth compression stroke, verify that the operational force does not change and that there is no unusual noise.
 - If there is any malfunction, replace the shock absorber.

FRONT SUSPENSION

FRONT SHOCK ABSORBER DISPOSAL

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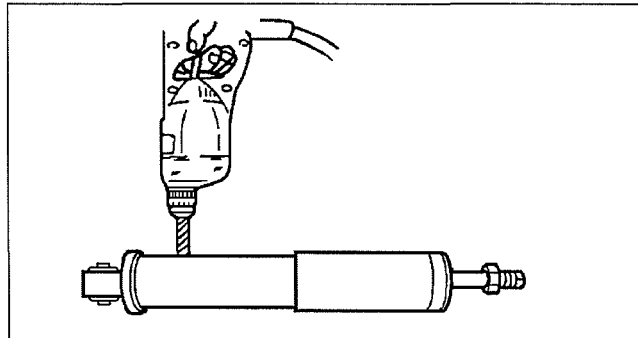
Warning

- Whenever drilling into a shock absorber, wear protective eye wear. The gas in the shock absorber is pressurized, and could spray metal chips into the eyes and face when drilling.

1. Clamp a shock absorber on a flat surface or with the piston pointing downwards.
2. Drill a 2—3 mm {0.08—0.12 in} hole at a point 20—30 mm {0.8—1.2 in} from the bottom of the tube, so that the gas can escape.
3. Turn the hole downwards.
4. The oil can be collected by moving the piston rod several times up and down and cutting the tube at the end.
5. Dispose of waste oil according to the waste disposal law.

Note

- Shock absorber gas is nitrogen gas.
- Shock absorber oil is mineral oil.

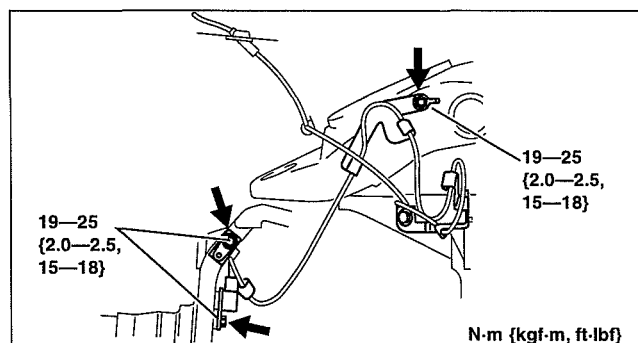


ABR7412W003

TORSION BAR SPRING AND LOWER ARM REMOVAL/INSTALLATION [4x2 (EXCEPT Hi-Rider)]

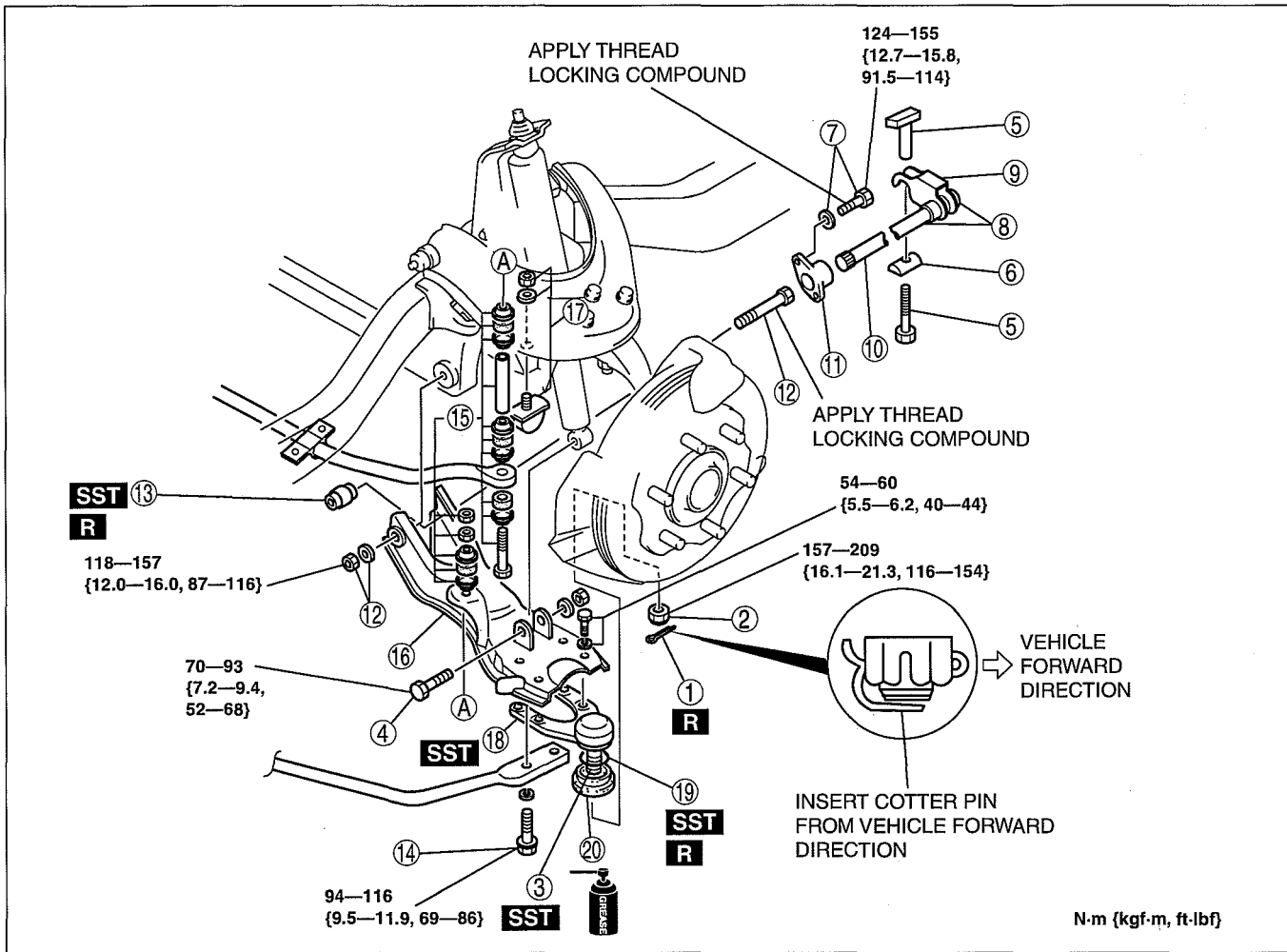
dcf021334130w01

1. Remove the ABS sensor wiring harness brackets installed to the upper arm and steering knuckle, then move aside the ABS sensor.
2. Remove in the order indicated in the table.
3. Install in the reverse order of removal.
4. Inspect the vehicle height and adjust it if necessary. (See 02-11-1 VEHICLE HEIGHT ADJUSTMENT [4x2].)
5. Inspect the front wheel alignment and adjust it if necessary. (See 02-11-3 FRONT WHEEL ALIGNMENT [4x2].)



DBG213ZWB033

FRONT SUSPENSION



DBG0213ZWB00

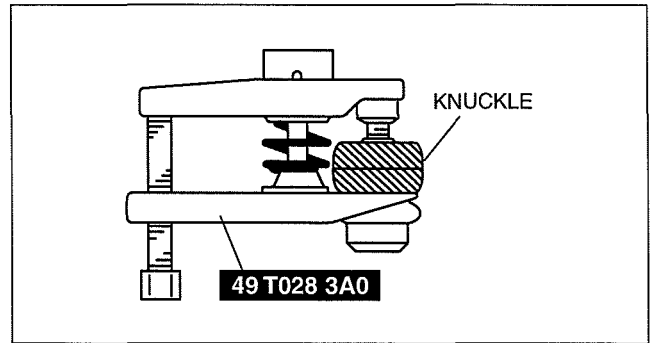
1	Cotter pin
2	Nut
3	Front lower arm ball joint (See 02-13-7 Front Lower Arm Ball Joint Removal Note.)
4	Shock absorber lower bolt and nut
5	Anchor bolt (See 02-13-7 Anchor Bolt Removal Note.) (See 02-13-9 Anchor Bolt Installation Note.)
6	Anchor swivel
7	Bolt and washer
8	Torsion bar component
9	Anchor arm
10	Torsion bar spring (See 02-13-7 Torsion Bar Spring Removal Note.) (See 02-13-8 Torsion Bar Spring Installation Note.)
11	Torque plate

12	Lower arm spindle, washer, and nut
13	Rubber bushing (See 02-13-7 Rubber Bushing Removal Note.) (See 02-13-8 Rubber Bushing Installation Note.)
14	Tension rod bolt
15	Stabilizer bolt, bushing, retainer, spacer, and nut (See 02-13-8 Stabilizer Bolt, Bushing, Retainer, Spacer, And Nut Installation Note.)
16	Front lower arm
17	Bound stopper
18	Front lower arm ball joint
19	Clip (See 02-13-8 Clip Installation Note.)
20	Dust boot
21	Clip (See 02-13-8 Clip Installation Note.)
22	Dust boot

FRONT SUSPENSION

Front Lower Arm Ball Joint Removal Note

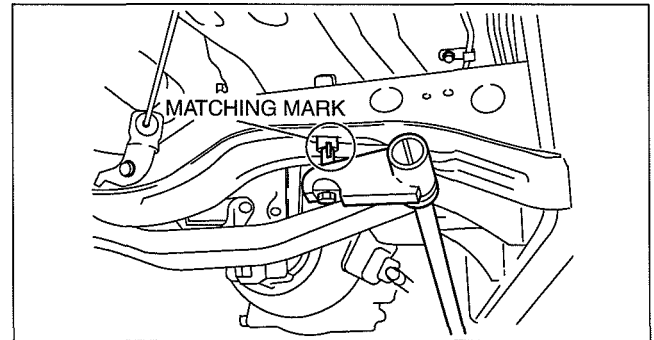
1. Separate the ball joint from the knuckle arm using the SST.



DBG0213ZWB00

Anchor Bolt Removal Note

1. Mark the anchor bolt and swivel for reference during installation.

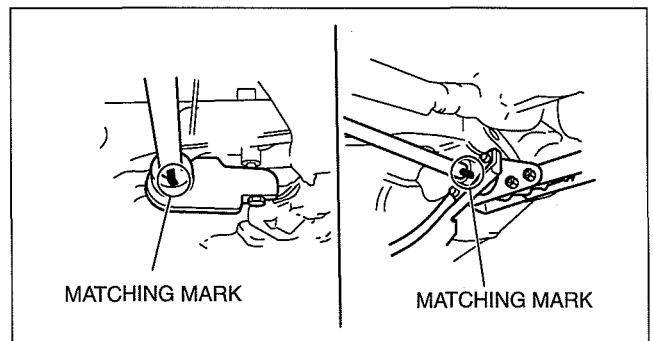


DBG0213ZWB00

02

Torsion Bar Spring Removal Note

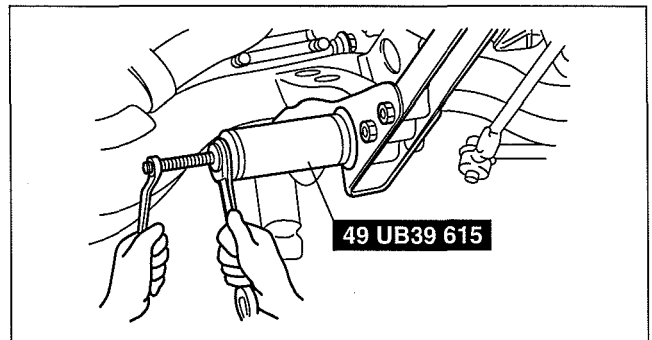
1. Mark the torsion bar spring, anchor arm, the torsion bar spring and torque plate for reference during installation.



DBG0213ZWB00

Rubber Bushing Removal Note

1. Remove the rubber bushing from the body using the SST.

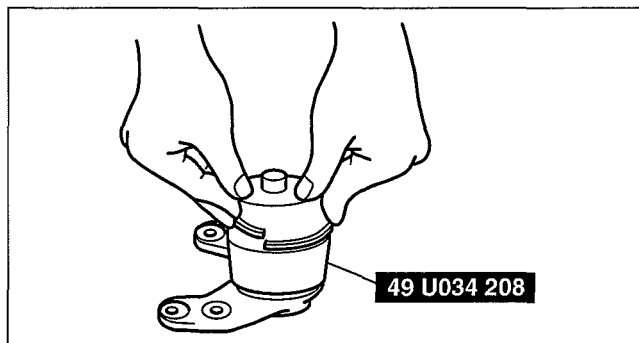


DBG0213ZWB00

FRONT SUSPENSION

Clip Installation Note

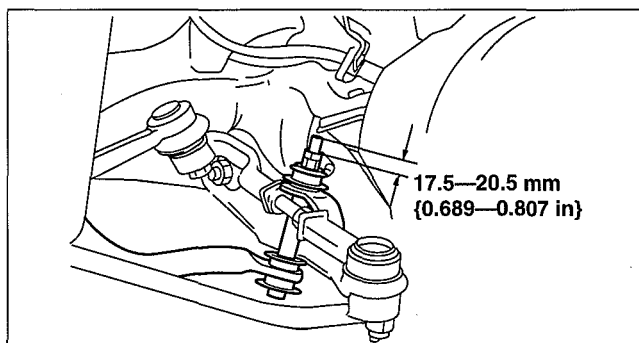
1. Install the **SST** to the ball joint stud with the stud stands straight up.
2. Install the clip in the dust boot groove.



DBG0213ZWB00

Stabilizer Bolt, Bushing, Retainer, Spacer, And Nut Installation Note

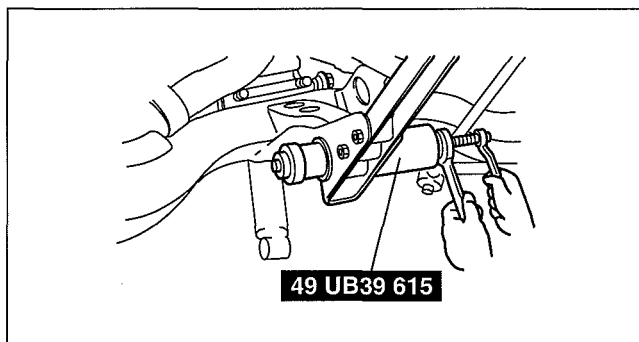
1. Tighten the nuts so that **17.5—20.5 mm {0.689—0.807 in}** of thread is exposed at the end of the bolt.



DBG0213ZWB00

Rubber Bushing Installation Note

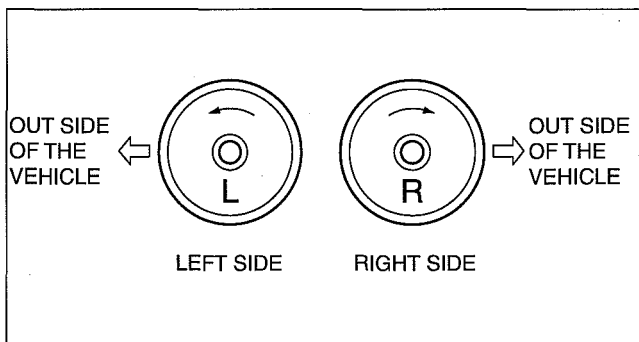
1. Install a new bushing using the **SST**.



DBG0213ZWB01

Torsion Bar Spring Installation Note

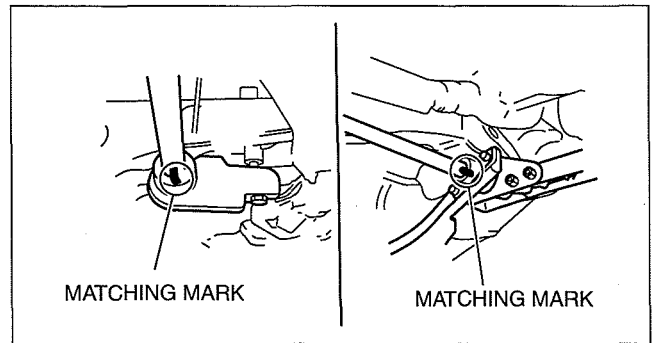
1. Before installation, check the identification mark on the end of the torsion bar spring.



DBG213ZWB031

FRONT SUSPENSION

- Align the marks made during removal, and connect the torsion bar spring to the torque plate.



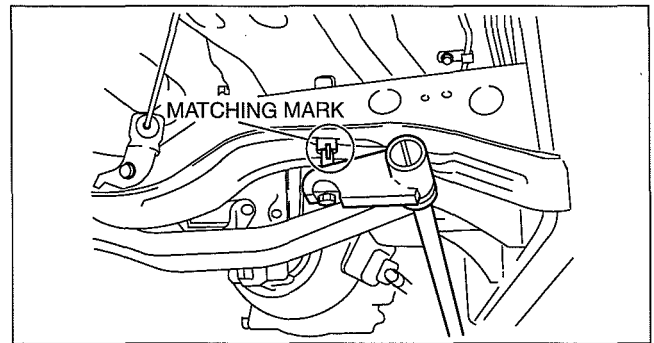
DBG0213ZWB00

Anchor Bolt Installation Note

- Install the anchor bolt, and tighten it until the marks made during removal are aligned.

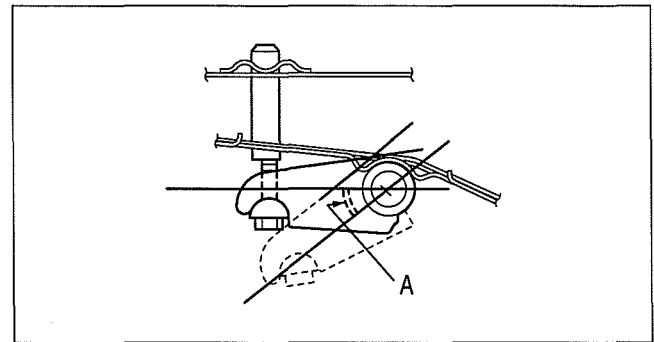
Note

- If the anchor bolt was not marked during removal, install it as follows:
 - Lower the front suspension until the upper arm contacts the rebound stopper.



DBG0213ZWB01

- Install the anchor arm so that the angle A is 25° — 34° .

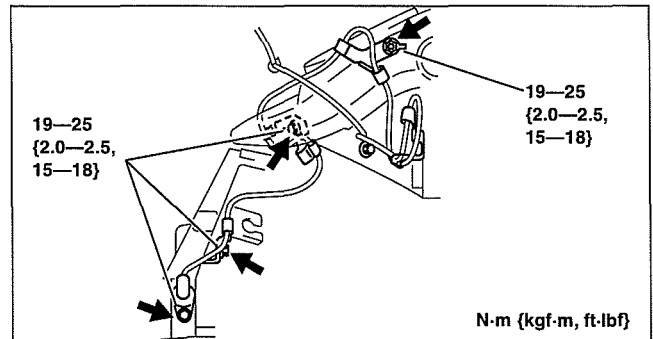


DBG0213ZWB01

TORSION BAR SPRING AND LOWER ARM REMOVAL/INSTALLATION [Hi-Rider, 4x4]

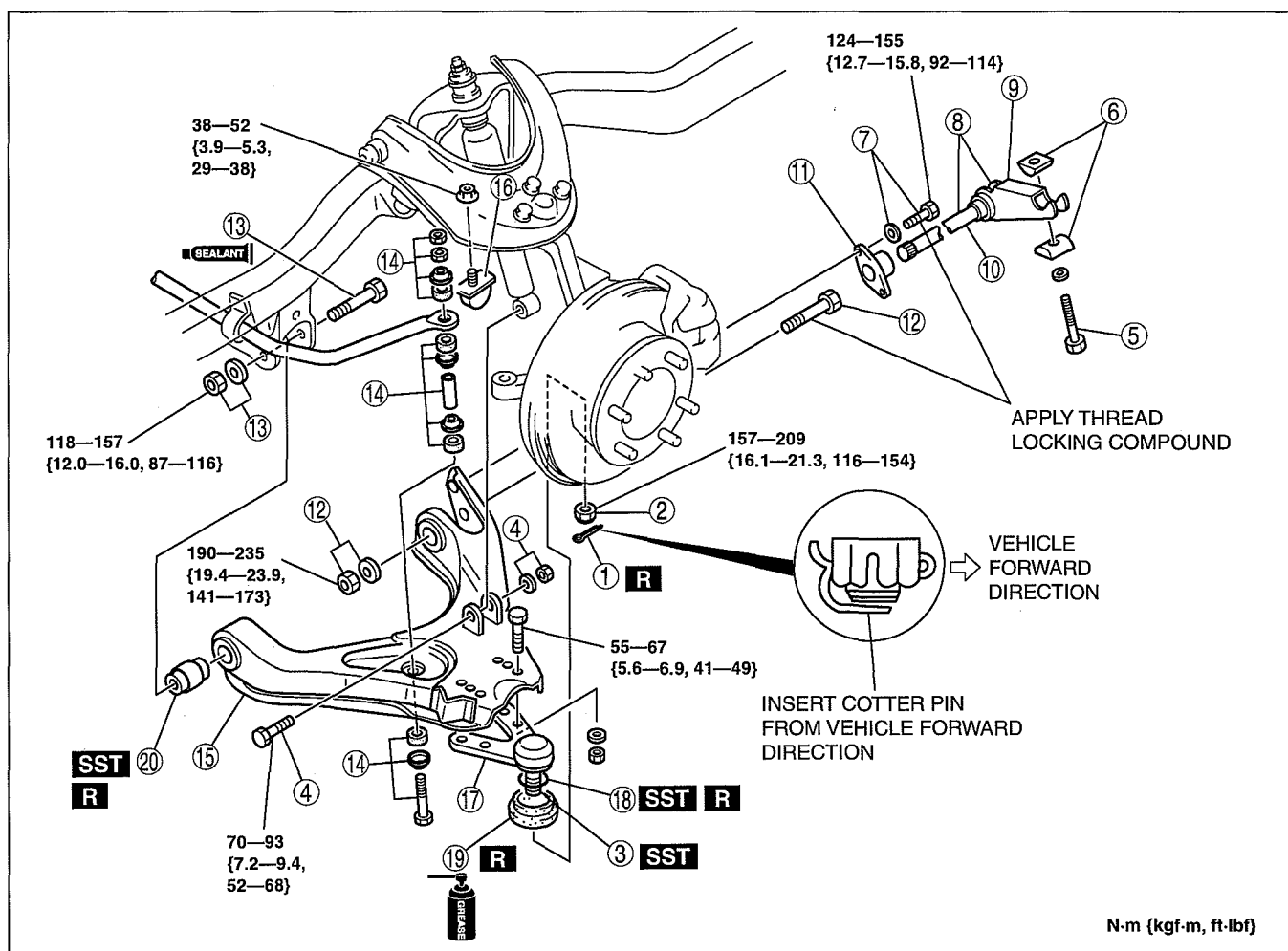
dcf021334130w02

- Remove the ABS sensor wiring harness brackets installed to the upper arm and steering knuckle, then move aside the ABS sensor.
- Remove in the order indicated in the table.
- Install in the reverse order of removal.
- Inspect the vehicle height and adjust it as necessary. (See 02-11-2 VEHICLE HEIGHT ADJUSTMENT [4x4].)
- Inspect the front wheel alignment as necessary. (See 02-11-5 FRONT WHEEL ALIGNMENT [4x4].)



DBG213ZWB034

FRONT SUSPENSION



DBG0213ZWB01

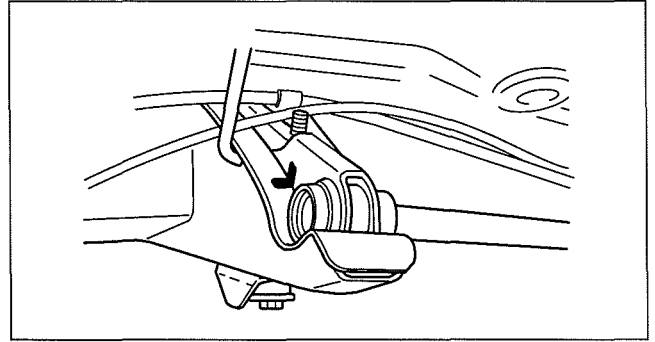
1	Cotter pin
2	Nut
3	Front lower arm ball joint (See 02-13-7 Front Lower Arm Ball Joint Removal Note.)
4	Shock absorber lower bolt and nut
5	Anchor bolt (See 02-13-13 Anchor Bolt Installation Note.)
6	Anchor swivel
7	Bolt and washer
8	Torsion bar spring component
9	Anchor arm (See 02-13-11 Anchor Arm Removal Note.) (See 02-13-12 Anchor Arm Installation Note.)
10	Torsion bar spring (See 02-13-11 Torsion Bar Spring Removal Note.) (See 02-13-12 Torsion Bar Spring Installation Note.)

11	Torque plate
12	Lower arm spindle (rear), washer and nut
13	Lower arm spindle (front), washer and nut
14	Stabilizer bolt, bushing, retainer, spacer and nut (See 02-13-8 Stabilizer Bolt, Bushing, Retainer, Spacer, And Nut Installation Note.)
15	Front lower arm
16	Bound stopper
17	Front lower arm ball joint
18	Clip (See 02-13-8 Clip Installation Note.)
19	Dust boot
20	Front lower arm bushing (See 02-13-11 Front Lower Arm Bushing Removal Note.) (See 02-13-12 Front Lower Arm Bushing Installation Note.)

FRONT SUSPENSION

Anchor Arm Removal Note

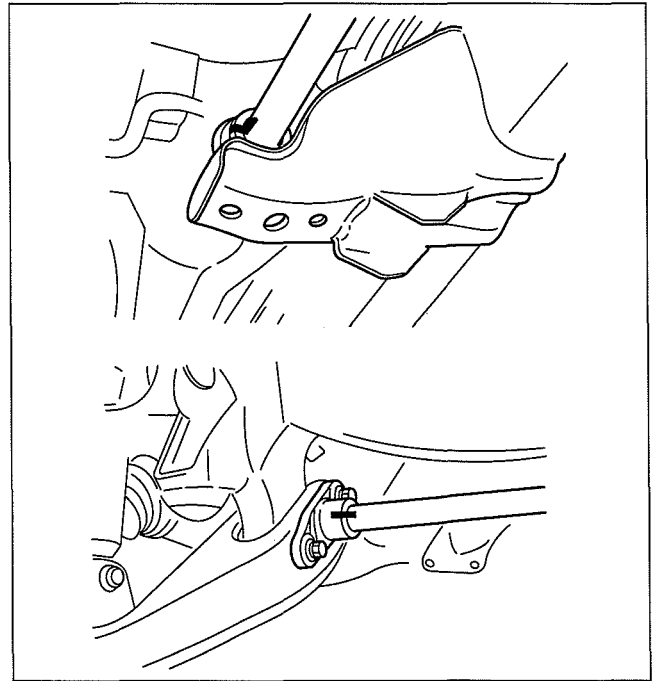
1. Mark the anchor arm and body for reference during installation.



DBG0213ZWB01

Torsion Bar Spring Removal Note

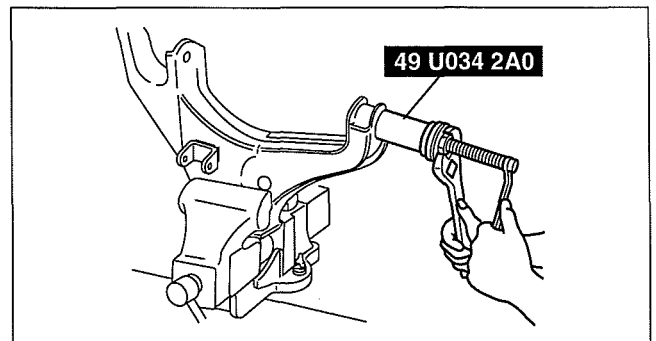
1. Mark the torsion bar spring and anchor arm, and torsion bar spring and torque plate for reference during installation.



DBG0213ZWB01

Front Lower Arm Bushing Removal Note

1. Remove the front lower arm bushing using the SST.

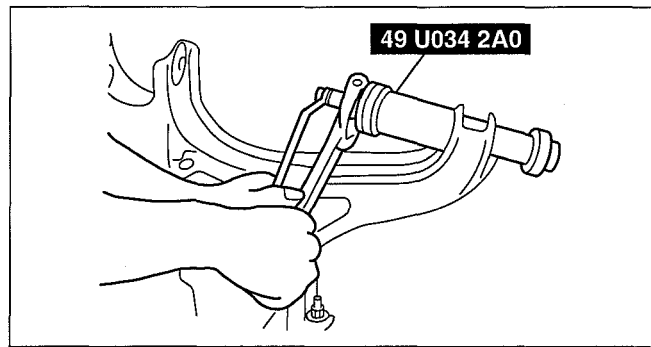


DBG0213ZWB01

FRONT SUSPENSION

Front Lower Arm Bushing Installation Note

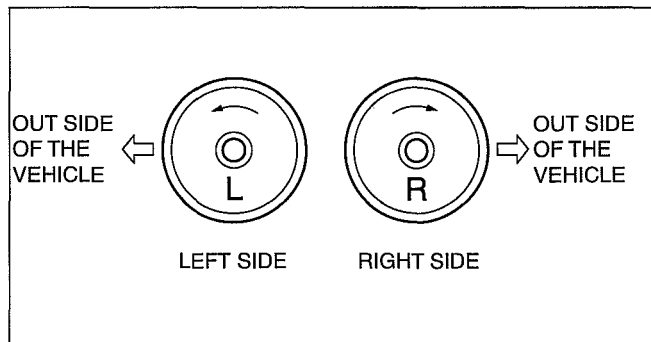
1. Apply soapy water to the new bushing.
2. Install the bushing using the **SST**.



DBG0213ZWB01

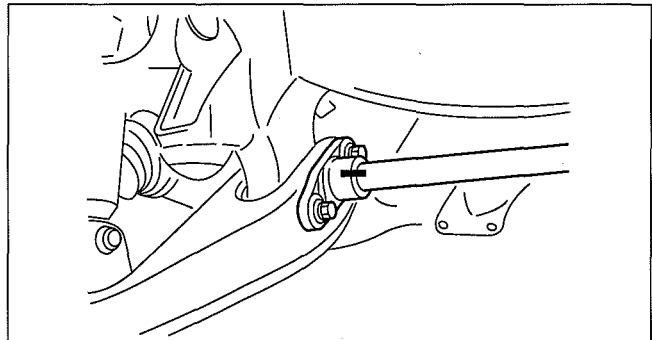
Torsion Bar Spring Installation Note

1. Before installation, check the identification mark on the end of the torsion bar spring.



DBG213ZWB031

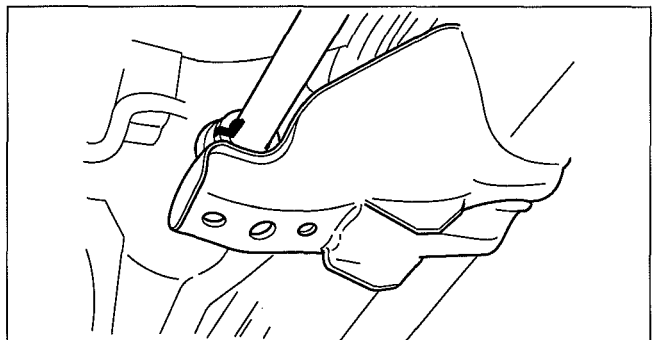
2. Align the marks made during removal, and connect the torsion bar spring to the torque plate.



DBG0213ZWB01

Anchor Arm Installation Note

1. Align the marks made during removal, and install the anchor arm onto the torsion bar spring.



DBG0213ZWB01

FRONT SUSPENSION

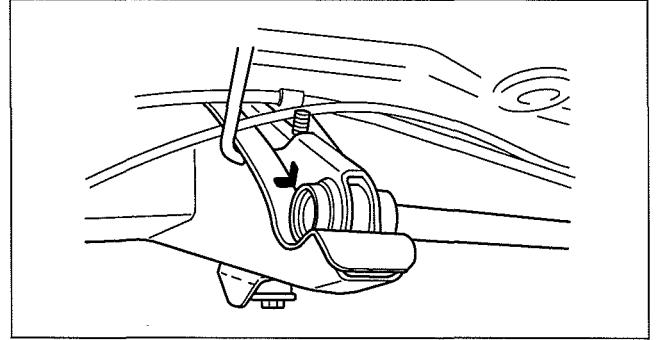
Anchor Bolt Installation Note

1. Install the anchor bolt, and tighten it until the marks made during removal are aligned.

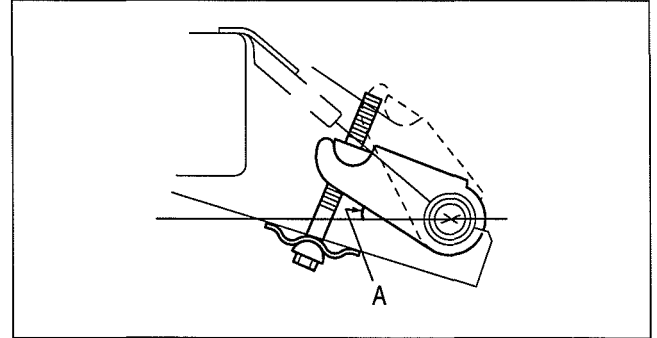
Note

- If the anchor bolt was not marked during removal, install it as follows:
 1. Lower the front suspension until the upper arm contacts the rebound stopper.

2. Install the anchor arm so that the angle A is $55^{\circ}30'$ — $64^{\circ}30'$.



DBG0213ZWB01



DBG0213ZWB02

02

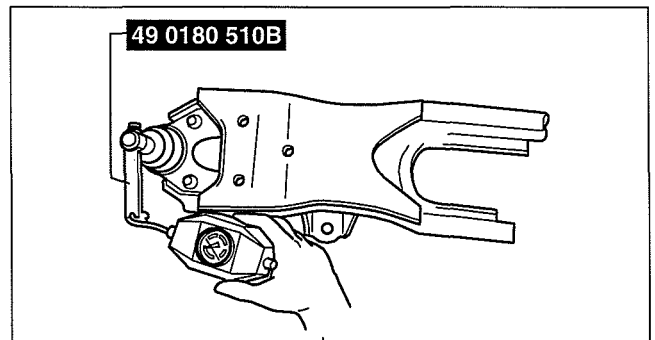
dcf021334130w03

FRONT LOWER ARM INSPECTION

1. Remove the front lower arm from the vehicle.
2. Inspect the front lower arm for bending or damage. If there is any malfunction, replace it.
3. Inspect the ball joint for excessive play. If there is any malfunction, replace the front lower arm.
4. Rotate the ball joint **5 times**.
5. Measure the ball-joint rotational torque using the SST and torque wrench.

Front lower arm ball joint rotational torque
[pull scale reading]
20—34 N {2.0—3.5 kgf, 4.4—7.7 lbf}

- If not within the specification, replace the front lower arm.



DBG0213ZWB02

FRONT SUSPENSION

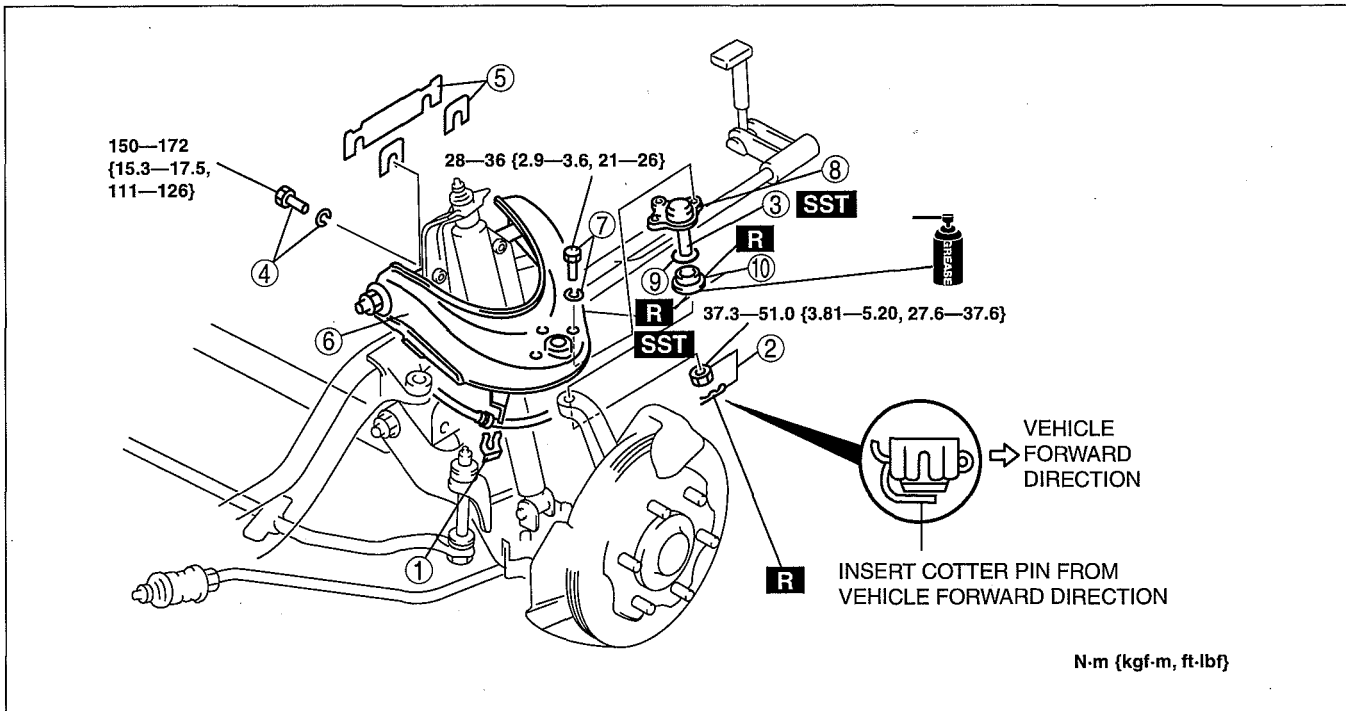
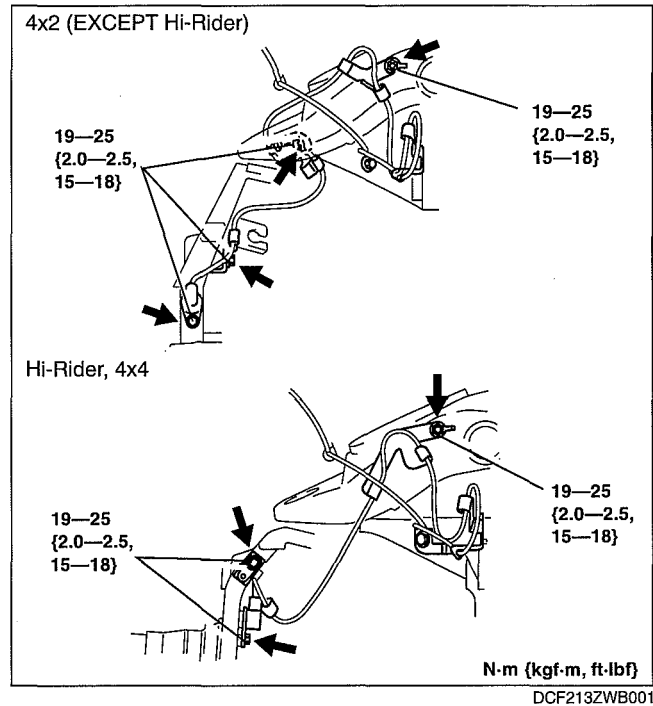
FRONT UPPER ARM REMOVAL/INSTALLATION

dcf021334200w01

Caution

- Performing the following procedures without first removing the ABS wheel-speed sensor may possibly cause an open circuit in the wiring harness if it is pulled by mistake. Before performing the following procedures, remove the ABS wheel-speed sensor (axle side) and fix it to an appropriate place where the sensor will not be pulled while servicing the vehicle.

- Remove the ABS sensor wiring harness brackets installed to the upper arm and steering knuckle, then move aside the ABS sensor.
- Remove in the order indicated in the table.
- Install in the reverse order of removal.
- Inspect the front wheel alignment and adjust it as necessary. (See 02-11-3 FRONT WHEEL ALIGNMENT [4x2]) (See 02-11-5 FRONT WHEEL ALIGNMENT [4x4].)



1	Brake hose bracket
2	Cotter pin and nut
3	Front upper arm ball joint (See 02-13-15 Front Upper Arm Ball Joint Removal Note.)

4	Bolt and washer (See 02-13-15 Bolt and Washer Installation Note.)
5	Adjustment shim (See 02-13-15 Adjustment Shim Removal Note.)
6	Front upper arm component
7	Bolt and washer

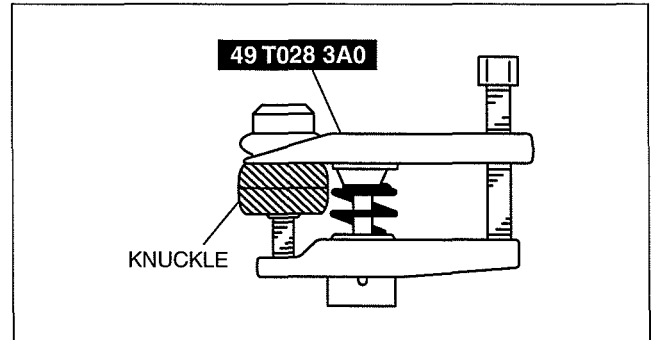
FRONT SUSPENSION

8	Front upper arm ball joint
---	----------------------------

9	Clip (See 02-13-15 Clip Installation Note.)
10	Dust boot

Front Upper Arm Ball Joint Removal Note

1. Separate the upper arm ball joint from the knuckle arm using the **SST**.



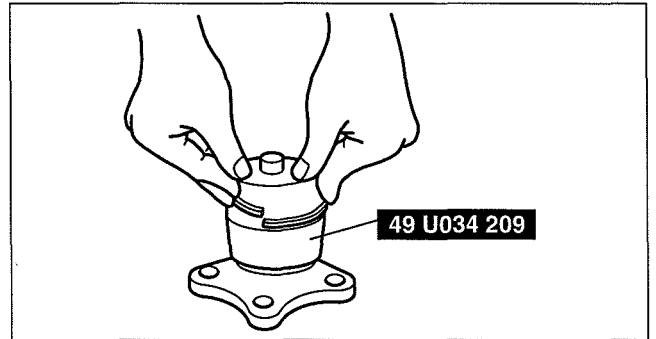
DBG0213ZWB02

Adjustment Shim Removal Note

1. Note the number, amount and position of the shims so that they are reinstalled in the correct positions.

Clip Installation Note

1. Install the **SST** to the ball joint stud with the stud stands straight up.
2. Install the clip in the dust boot groove using the **SST**.



DBG0213ZWB02

Bolt and Washer Installation Note

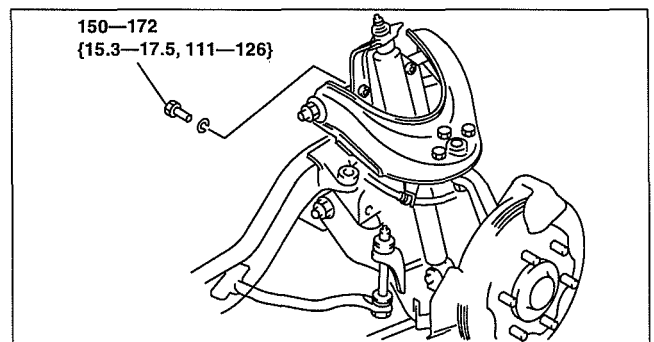
Note

- Be sure to use a pre-set torque wrench to precisely tighten the upper arm shaft.

1. Install the bolt and washer, and tighten the bolt to the specified torque.

Tightening torque

150—172 N·m {15.3—17.5 kgf·m, 111—126 ft·lbf}



DCF213ZWB300

FRONT SUSPENSION

FRONT UPPER ARM INSPECTION

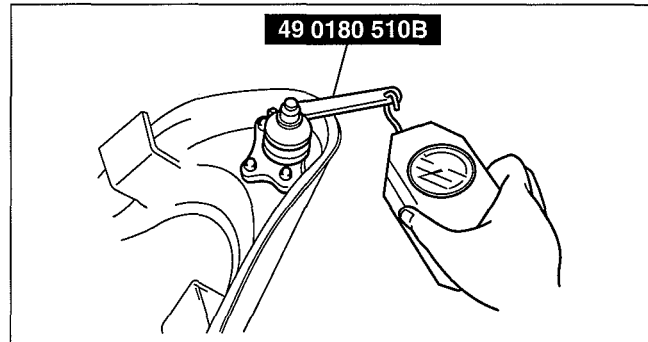
dcf021334200w02

1. Remove the front upper arm from the vehicle.
2. Inspect the front upper arm for bending or damage. If there is any malfunction, replace it.
3. Inspect the ball joint for excessive play, and if there is any malfunction, replace the front upper arm.
4. Rotate the ball joint **5 times**.
5. Measure the ball-joint rotational torque using the **SST** and pull scale.

Front upper arm ball joint rotational torque
[pull scale reading]

4.9—25 N {0.5—2.5 kgf, 1.1—5.5 lbf}

- If not within the specification, replace the front upper arm.

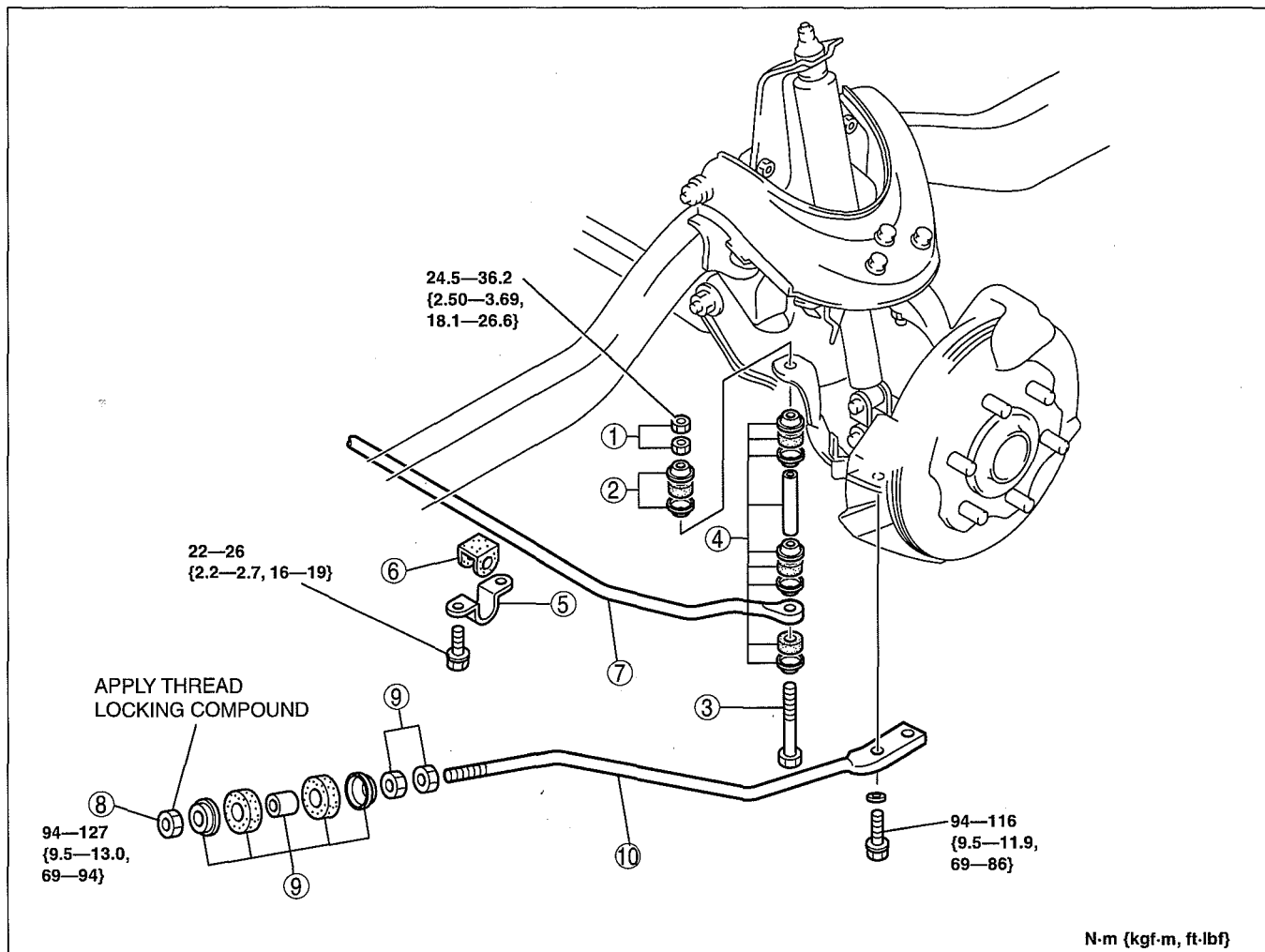


DBG0213ZW02

FRONT STABILIZER AND TENSION ROD REMOVAL/INSTALLATION [4x2 (EXCEPT Hi-Rider)]

dcf021334100w01

1. Remove in the order indicated in the table.
2. Install in the reverse order of removal.



DBG0213ZW02

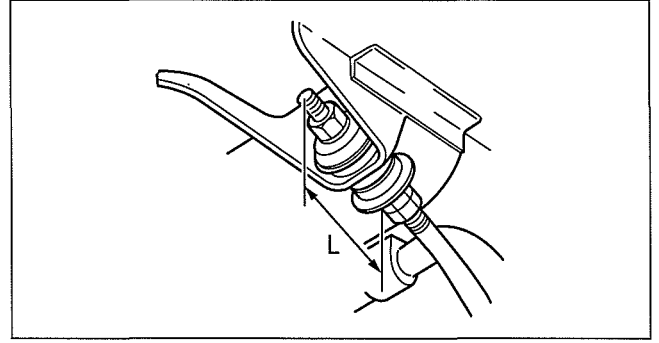
1	Nut (See 02-13-8 Stabilizer Bolt, Bushing, Retainer, Spacer, And Nut Installation Note.)
2	Retainer
3	Bolt
4	Bushing, retainer and spacer
5	Stabilizer bracket

6	Stabilizer bushing
7	Front stabilizer
8	Tension rod nut (See 02-13-17 Tension Rod Nut Installation Note.)
9	Bushing and retainer
10	Tension rod

FRONT SUSPENSION

Tension Rod Nut Installation Note

1. Tighten the tension rod nuts so that length L is 98.5 mm {3.88 in}.

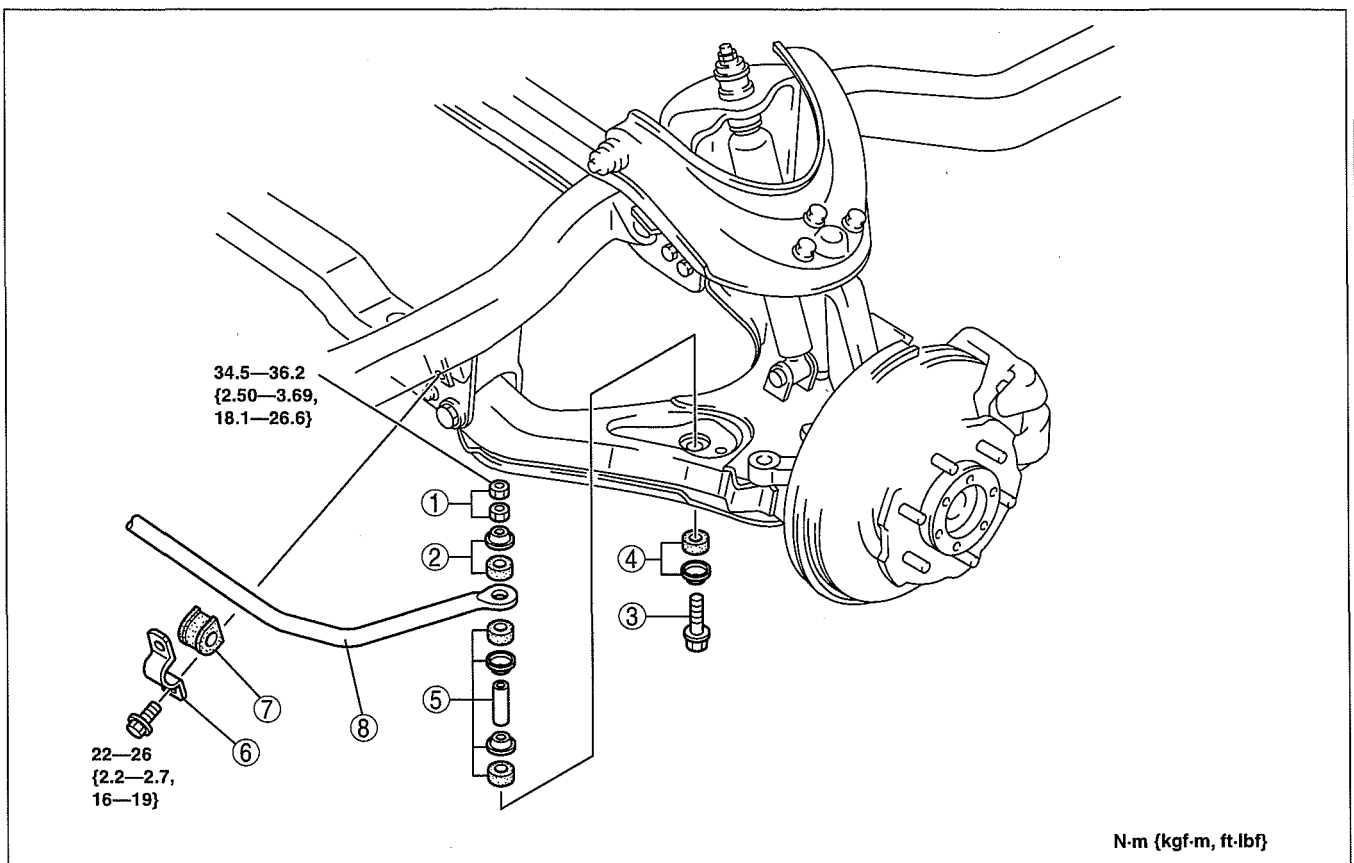


DBG0213ZWB02

FRONT STABILIZER REMOVAL/INSTALLATION [Hi-Rider, 4x4]

dcf021334100w02

1. Remove in the order indicated in the table.
2. Install in the reverse order of removal.



DBG0213ZWB02

1	Nut (See 02-13-8 Stabilizer Bolt, Bushing, Retainer, Spacer, And Nut Installation Note.)
2	Retainer and bushing
3	Bolt

4	Retainer and bushing
5	Retainer, bushing, and spacer
6	Stabilizer bracket
7	Stabilizer bushing
8	Front stabilizer

REAR SUSPENSION

02-14 REAR SUSPENSION

REAR SUSPENSION

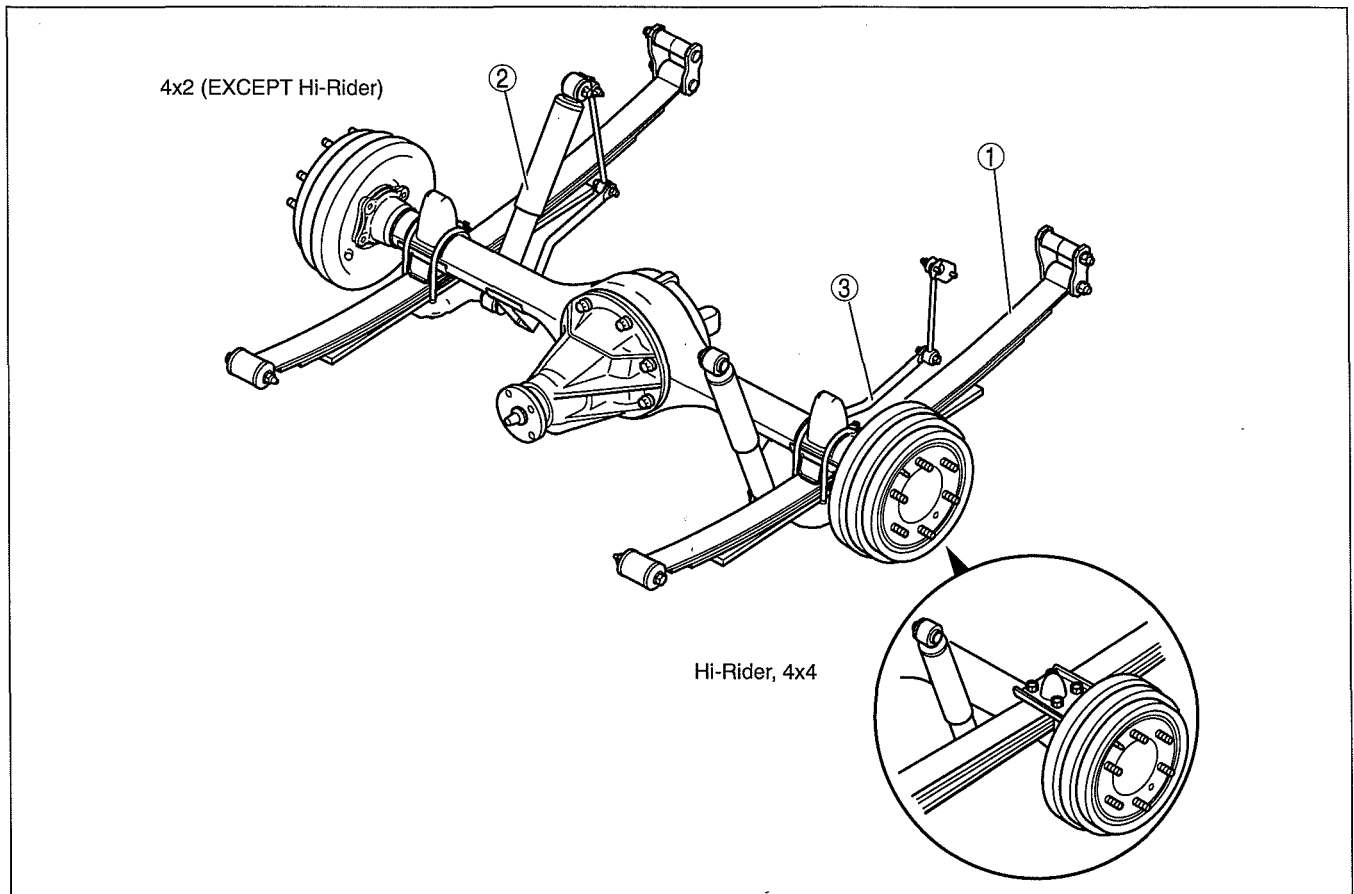
LOCATION INDEX.....	02-14-1
REAR SHOCK ABSORBER AND LEAF SPRING REMOVAL/INSTALLATION [4x2 (EXCEPT Hi-Rider)].....	02-14-2
REAR SHOCK ABSORBER AND LEAF SPRING REMOVAL/INSTALLATION [Hi-Rider, 4x4]	02-14-3

REAR SHOCK ABSORBER

INSPECTION.....	02-14-4
REAR SHOCK ABSORBER DISPOSAL.....	02-14-4
REAR STABILIZER REMOVAL/ INSTALLATION [WL-C, WE-C]	02-14-5

REAR SUSPENSION LOCATION INDEX

dcf02140000w01



DCF214ZWB001

1	Leaf spring (See 02-14-2 REAR SHOCK ABSORBER AND LEAF SPRING REMOVAL/INSTALLATION [4x2 (EXCEPT Hi-Rider)].) (See 02-14-3 REAR SHOCK ABSORBER AND LEAF SPRING REMOVAL/INSTALLATION [Hi- Rider, 4x4].)
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2	Rear shock absorber (See 02-14-4 REAR SHOCK ABSORBER INSPECTION.) (See 02-14-4 REAR SHOCK ABSORBER DISPOSAL.)
3	Rear stabilizer (WL-C and WE-C) (See 02-14-5 REAR STABILIZER REMOVAL/ INSTALLATION [WL-C, WE-C].)

REAR SUSPENSION

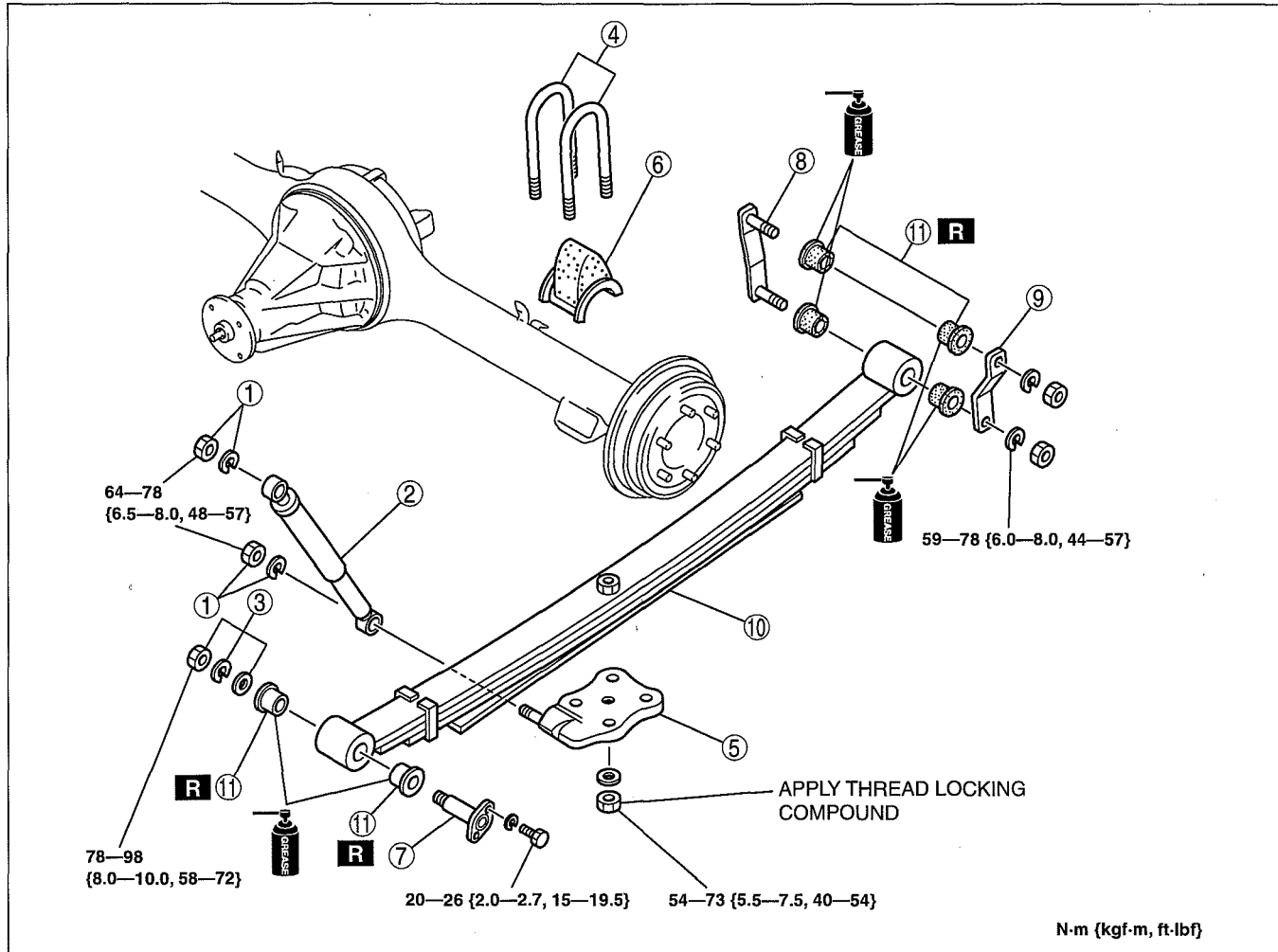
REAR SHOCK ABSORBER AND LEAF SPRING REMOVAL/INSTALLATION [4x2 (EXCEPT Hi-Rider)]

dcf021405910w01

Warning

- Use safety stands for support of the left and right axle casings, and while supporting the differential casing with a jack, raise or lower the jack as necessary.

1. Support the frame with safety stands and place a jack under the differential casing.
2. Remove in the order indicated in the table.
3. Install in the reverse order of removal.
4. After installation, tighten the bolt (leaf spring front side) completely on the ground.



DBG214ZWB001

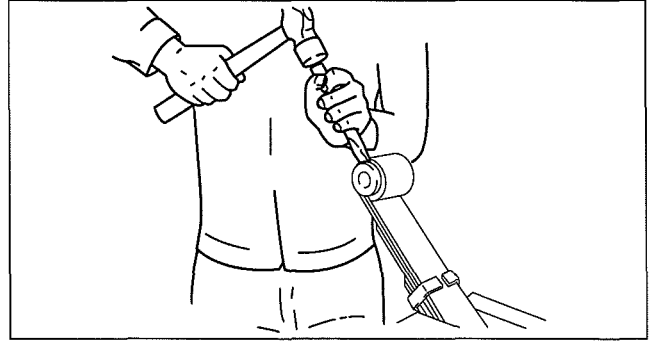
1	Nut and washer
2	Rear shock absorber
3	Nut
4	U-bolt
5	Spring clamp
6	Stopper rubber
7	Bolt

8	Shackle pin
9	Shackle plate
10	Leaf spring component
11	Leaf spring bushing (See 02-14-3 Leaf Spring Bushing Removal Note.) (See 02-14-3 Leaf Spring Bushing Installation Note.)

REAR SUSPENSION

Leaf Spring Bushing Removal Note

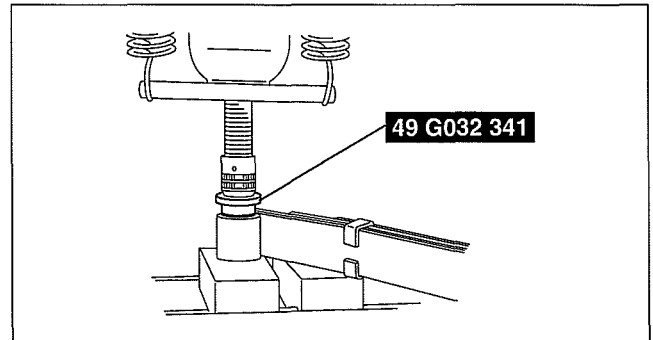
1. Secure the leaf spring in a vise and remove the bushing using a chisel.



AVF7416W004

Leaf Spring Bushing Installation Note

1. Press in the bushing using a suitable pipe.



DBG214ZWB100

02

REAR SHOCK ABSORBER AND LEAF SPRING REMOVAL/INSTALLATION [Hi-Rider, 4x4]

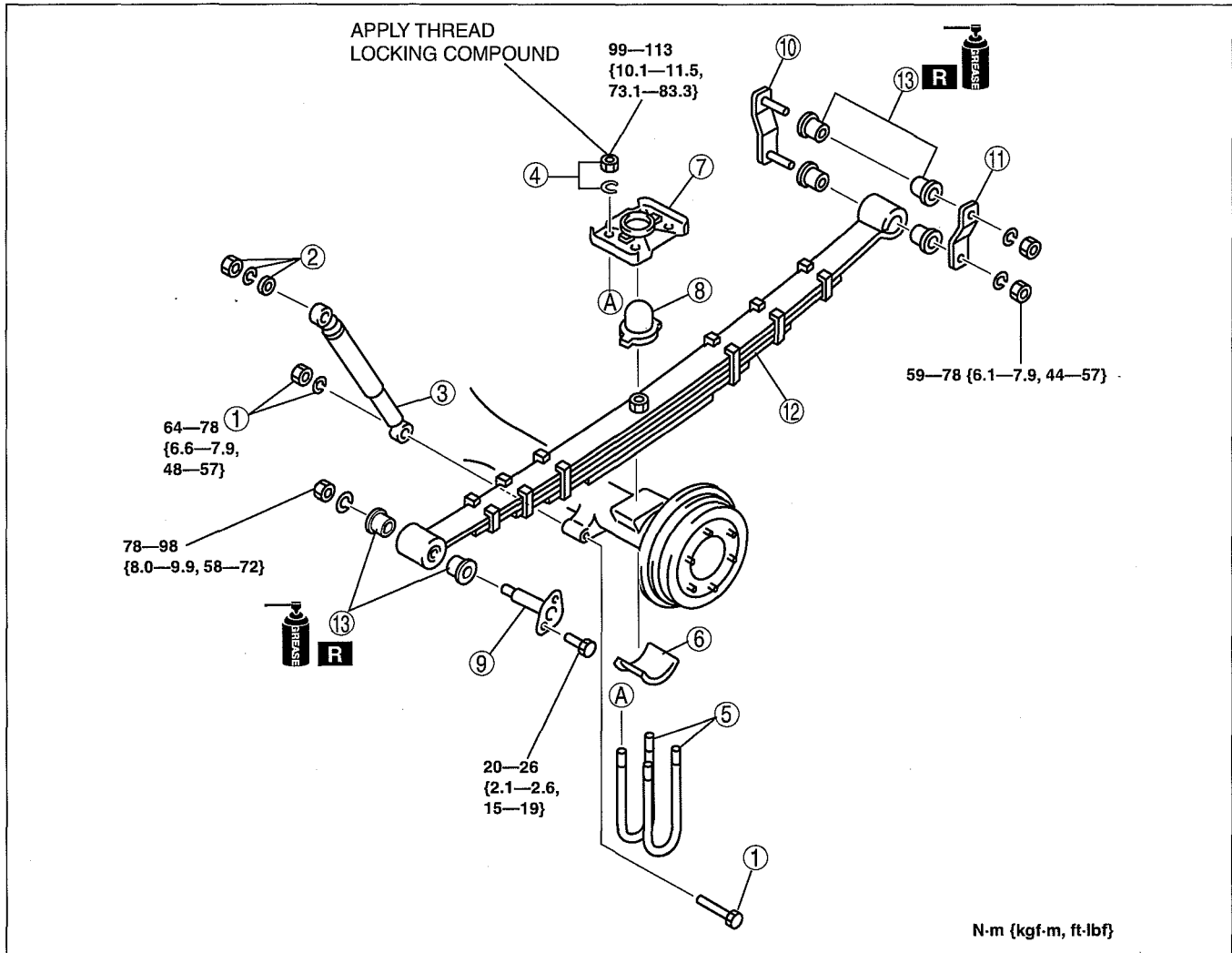
dcf021405910w02

Warning

- Use safety stands for support of the left and right axle casings, and while supporting the differential casing with a jack, raise or lower the jack as necessary.

1. Support the frame with safety stands and place a jack under the differential casing.
2. Remove in the order indicated in the table.
3. Install in the reverse order of removal.
4. After installation, tighten the bolt (leaf spring front side) completely on the ground.

REAR SUSPENSION



DBG214ZWB006

1	Bolt, washer and nut
2	Nut, washer, and retainer
3	Shock absorber
4	Nut and washer
5	U-bolt
6	Set plate
7	Spring clamp
8	Bound stopper

9	Bolt
10	Shackle pin
11	Shackle plate
12	Leaf spring component
13	Leaf spring bushing (See 02-14-3 Leaf Spring Bushing Removal Note.) (See 02-14-3 Leaf Spring Bushing Installation Note.)

REAR SHOCK ABSORBER INSPECTION

dcf021428700w01

1. Inspect the rear shock absorber in the same way as the front shock absorber.
(See 02-13-4 FRONT SHOCK ABSORBER INSPECTION.)

REAR SHOCK ABSORBER DISPOSAL

dcf021428700w02

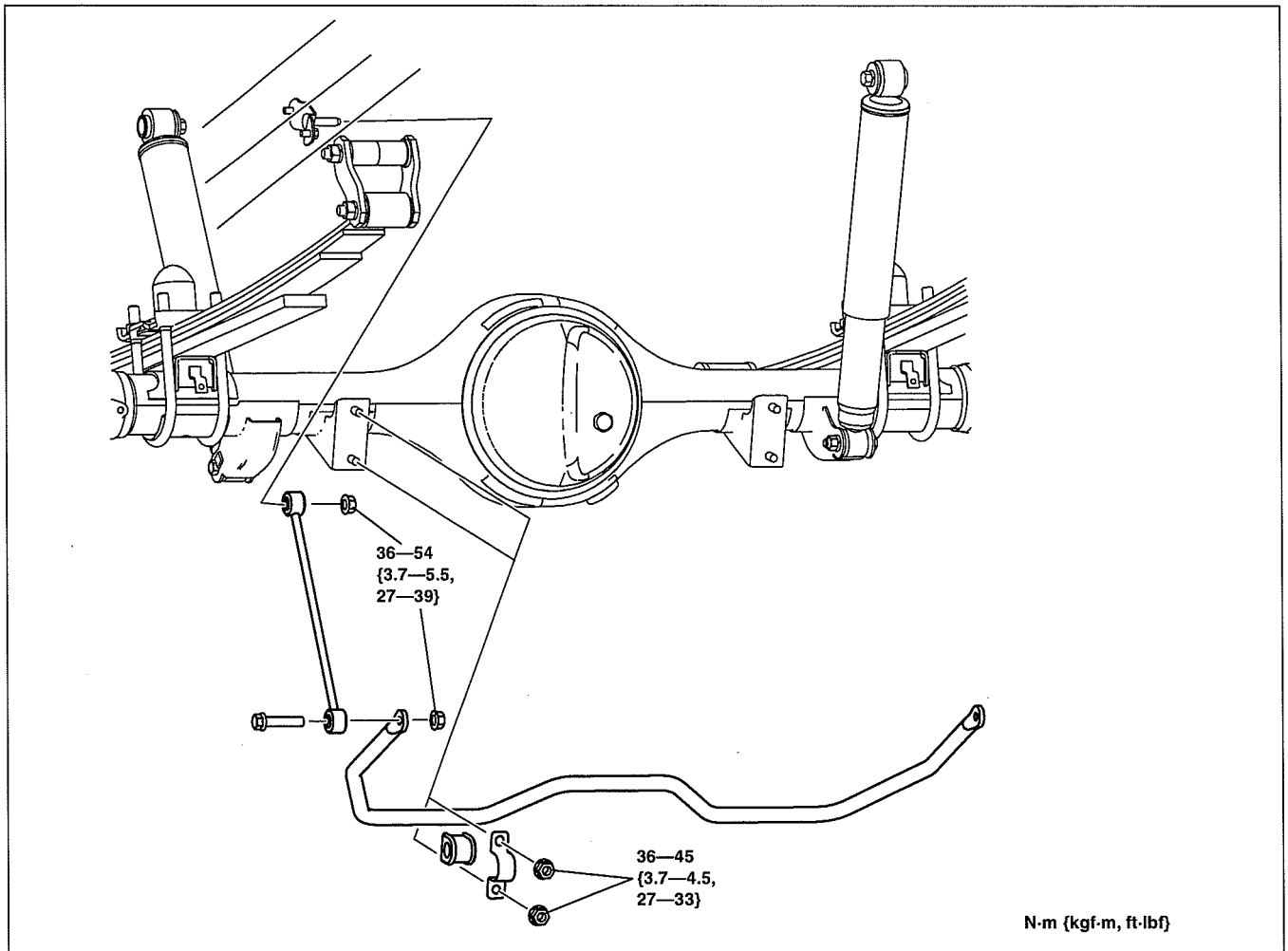
1. Dispose the rear shock absorber in the same way as the front shock absorber. (See 02-13-5 FRONT SHOCK ABSORBER DISPOSAL.)

REAR SUSPENSION

REAR STABILIZER REMOVAL/INSTALLATION [WL-C, WE-C]

dcf021428100w01

1. Remove in the order indicated in the table.
2. Install in the reverse order of removal.
3. After installation, tighten the control link installation bolt completely on the ground.



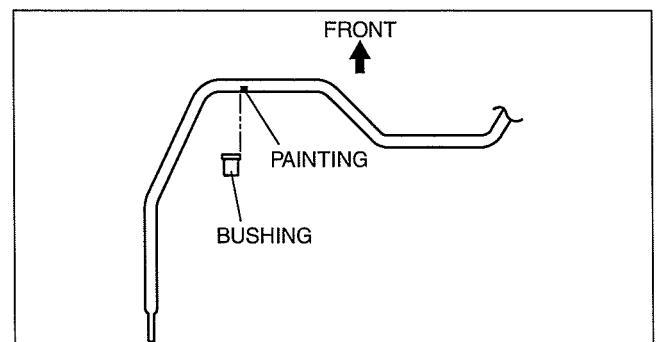
DBG214ZWB003

1	Rear stabilizer component (See 02-14-6 Rear Stabilizer Component Installation Note.)
2	Stabilizer control link (See 02-14-6 Stabilizer Control Link Installation Note.)

3	Stabilizer bracket
4	Bushing (See 02-14-5 Bushing Installation Note.)
5	Rear stabilizer

Bushing Installation Note

1. Install the bushings aligned with the painting part.



DBG214ZWB004

REAR SUSPENSION

Stabilizer Control Link Installation Note

1. Install the stabilizer control link with the painted area pointing towards the under side and inside of the vehicle.

Rear Stabilizer Component Installation Note

1. Temporarily install the rear stabilizer component.
2. Tighten the stabilizer control link nut.
3. Tighten the stabilizer bracket nut.

TECHNICAL DATA

02-50 TECHNICAL DATA

SUSPENSION TECHNICAL DATA 02-50-1

SUSPENSION TECHNICAL DATA

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Front wheel alignment [4x2] (Unloaded)

Item			Specifications			
			Regular cab	Double cab	Hi-Rider (Stretch cab (with Rear Access System))	Hi-Rider (Double cab)
Total toe-in	Tire [Tolerance ± 3 {0.12}]	(mm {in})	5 {0.19}			
Maximum steering angle		Inner	33°00'—37°00'		31°30'—35°30'	
		Outer	30°00'—35°00'		27°00'—32°00'	
Camber angle [Tolerance $\pm 1^\circ$]			0°35'		0°44'	0°45'
Caster angle [Tolerance $\pm 1^\circ$]			1°56'	1°54'	2°07'	2°06'
Steering axis inclination (reference value)			8°25'		10°37'	10°35'

Note

- Unloaded vehicle: Fuel tank is full. Engine coolant and engine oil are at specified level. Jack and tools are in designated position.
- Difference between the left and right dimension for camber is within 30'.
- Difference between the left and right dimension for caster is within 45'.

Front wheel alignment [4x4] (Unloaded)

Item			Specifications	
			Stretch cab (with Rear Access System)	Double cab
Total toe-in	Tire [Tolerance ± 3 {0.12}]	(mm {in})	6 {0.23}	
Maximum steering angle		Inner	31°30'—35°30'	
		Outer	27°00'—32°00'	
Camber angle (reference value) [Tolerance $\pm 1^\circ$]			0°44'	0°45'
Caster angle (reference value) [Tolerance $\pm 1^\circ$]			2°07'	2°06'
Steering axis inclination (reference value)			10°37'	10°35'

Note

- Unloaded vehicle: Fuel tank is full. Engine coolant and engine oil are at specified level. Jack and tools are in designated position.
- Difference between the left and right dimension for camber is within 30'.
- Difference between the left and right dimension for caster is within 45'.

TECHNICAL DATA

Wheel and tires [Australian Specs.]

Item				Specification			
Wheel	Size			15 × 6 1/2J		15 × 6 1/2J	16 × 7J
	Offset (mm {in})			20 {0.79}		25 {0.98}	10 {0.39}
	Pitch circle diameter (mm {in})			139.7 {5.500}			
	Material			Steel	Aluminum alloy	Steel	Aluminum alloy
Tire	Size			215/70R15C 106/104S		235/75R15 109S	245/70R16 111S
	Air pressure (kPa {(kgf/cm ²), <psi>})	Front	Up to full persons	220 {(2.2), <32>}		210 {(2.1), <30>}	210 {(2.1), <30>}
			Full load				
		Rear	Up to full persons	210 {(2.1), <30>}	210 {(2.1), <30>}	210 {(2.1), <30>}	
			Full load	375 {(3.8), <54>}	290 {(2.9), <42>}	270 {(2.7), <39>}	
	Remaining tread (mm {in})			1.6 {0.06}			
Wheel and tire	Lug nut tightening torque (N?m {kgf?m, ft?lbf})			88.2—117.6 {9.00—11.99, 65.06—86.73}			
	Wheel and tire runout (mm {in})		Radial direction	1.5 {0.06} max.			
			Lateral direction	2.0 {0.08} max.			
	Wheel imbalance*2 (g {oz})			10 {0.35} max.			

*1 : Total weight exceeds **120 g {4.24 oz}**.

*2 : One balance weight: **60 g {2.12 oz}** max. If the total weight exceeds **120 g {4.24 oz}** on one side, rebalance after moving the tire around on the rim. Do not use three or more balance weights.

Suspension

Item	Specification
Front lower arm ball joint rotational torque [pull scale reading]	20—34 N {2.0—3.5 kgf, 4.4—7.7 lbf}
Front upper arm ball joint rotational torque [pull scale reading]	4.9—25 N {0.5—2.5 kgf, 1.1—5.5 lbf}

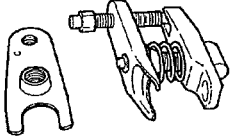
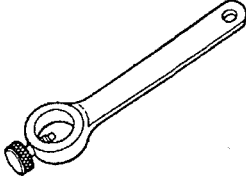
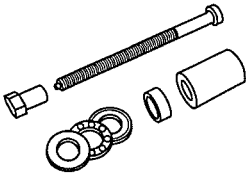
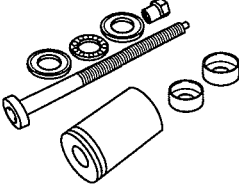
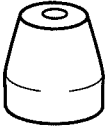
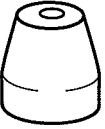
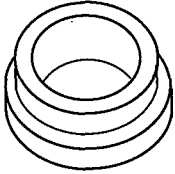
SERVICE TOOLS

02-60 SERVICE TOOLS

SUSPENSION 02-60-1

SUSPENSION

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49 T028 3A0 Ball joint puller set 	49 0180 510B Preload measuring attachment 	49 U034 2A0 Lower arm bushing puller & installer set 
49 UB39 615 Bushing puller & installer set 	49 U034 208 Clip guide 	49 U034 209 Clip guide 
49 G032 341 Oil seal installer 		

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